NED University of Engineering and Technology
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GPU Sharing for Multi-Tenant Inference

A Bounded Systematic Evidence Synthesis of Performance, Isolation and Engineering Trade-offs

Operating Systems CEP – Topic 23
Task 3: First Complete Draft

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Based on the supplied Task 2 corpus: 21 report versions, 20 unique studies.
The bounded selection and remaining verification work are explicitly reported.

Abstract

GPU sharing for multi-tenant inference requires a choice among resource efficiency, request latency, isolation and implementation cost. This bounded review synthesizes 20 studies represented by 21 reports using the supplied search, screening, appraisal and extraction records. The register contains 4,189 occurrences and 4,075 canonical records after 114 duplicate links. It records 3,493 screening exclusions, 559 retained records not advanced under the stopping rule, and 23 source-assessed reports, of which two are excluded. The analysis classifies temporal scheduling, software spatial sharing, hardware partitions, runtime/API mediation and hybrid controls. It uses 200 appraisal items, 52 reported observations and two transparent calculations. The evidence supports conditional selection: increased concurrency can improve throughput while increasing latency; foreground protection can reduce background progress; and higher model density can introduce transfer, compatibility or trust costs. No pooled effect or universal winning mechanism is supported. Recommendations are linked to source conditions and remain limited by bounded selection, title-only screening, a single assessment pass and source-version uncertainty.

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1 Introduction

1.1 Background and engineering problem

GPU sharing for inference is an operating-system resource-management problem: several services compete for finite execution capacity, device memory and access to a runtime that may not expose the control required by a service-level objective. Increasing concurrent execution can improve aggregate work completed while worsening the response of an individual service. Conversely, reserving the entire accelerator for one model can simplify interference management but leaves less opportunity to use spare capacity. The engineering question is therefore conditional: which mechanism should be selected for a particular workload, latency objective, memory requirement and isolation boundary?

The evidence makes this tension concrete. The ParvaGPU manuscript reports a local configuration in which throughput increases from 354 to 446 requests/s as the number of MPS processes rises from one to three, while reported latency increases from 11 to 27 ms. These outcomes move in opposite desirable directions. The latency quantile is unspecified, so the example cannot establish a P99 service guarantee. XSched instead reports a foreground P99 ratio of 1.07 relative to standalone execution, compared with 1.53 for native Triton in the recorded configuration. These are different engineering objectives, not numbers on one ranking scale [1, 2].

1.2 Scope and contribution

The unit of technical interpretation is concurrent inference on one physical GPU. A multi-GPU or cluster paper contributes only where a local sharing result can be separated from fleet scheduling. Mixed inference/training papers contribute inference outcomes and the associated background-work cost. Training throughput is not relabeled as inference throughput. Non-GPU accelerator results in otherwise relevant papers are outside the primary comparison.

This first draft synthesizes the supplied bounded corpus of 20 studies represented by 21 reports. Its contribution is a decision-oriented comparison that joins mechanisms, evaluation conditions, reported outcomes, implementation costs and limitations. It distinguishes performance interference, resource partitioning, memory protection and fault containment. It also retains adverse outcomes rather than selecting only favorable headline speedups.

The review is not an exhaustive census of the search results. The supplied record contains 559 retained records that were not advanced to source assessment. The first draft therefore supports candidate selection under observed conditions, while leaving the wider coverage question open. This boundary is part of the research method and must remain visible in the abstract, selection accounting and recommendations.

1.3 Objectives

The objectives are to classify the available mechanisms by their actual control boundaries; determine which outcomes are comparable; explain the conditions behind benefits and regressions; and derive recommendations that identify both expected benefit and sacrifice.

The review does not seek a universal winning system. It seeks an auditable reason for choosing a candidate under stated constraints.

2 Research questions

The following research questions reproduce the submitted Task 1 protocol (Table 3).

RQ1 What time-slicing, MPS, MIG and API remoting mechanisms are evaluated for concurrent inference on one GPU, and how do sharing granularity, layer and isolation boundary classify them?

RQ2 How do studies measure utilization, throughput, density, average/tail latency, predictability and interference, and how comparable are their experimental methods?

RQ3 Under what GPU, workload, batch, tenant mix and configuration conditions are changes in utilization, throughput, density and latency reported?

RQ4 What resource and fault isolation properties, trade-offs, implementation costs and conflicting results are reported, and what limits the evidence?

RQ5 Which approach or combination is supported for specified inference-serving conditions and priorities, with what confidence and alternatives?

3 Research methodology

3.1 Protocol and implemented method

The submitted Task 1 protocol is preserved unchanged in the repository. It specifies independent screening and appraisal, stable record/report/study identifiers, page-level extraction and a condition-aware narrative synthesis. This draft follows its conceptual comparison rules while explicitly reporting deviations in execution. PRISMA supplies the reporting framework, while the search audit preserves the executed queries rather than replacing them with planned expressions [3, 4, 5].

The supplied Task 2 workbook is the source of the current corpus and historical decisions. GitHub's preceding status described an earlier 15-form assessment. Importing the newer workbook does not mean those older registers were silently rewritten. Task 3 stores the uploaded baseline separately and keeps the inherited research history available.

The implemented review used a single consolidated AI-assisted assessment under team direction. The uploaded materials refer to team checking, but contain no named independent item-level votes or signed verification record. This draft therefore does not claim completion of the protocol's independent second review or report inter-rater agreement. New activity logs identify the AI assistant's actions and actual checkpoint times. Historical batch or workbook-build times are not evidence of when a person screened a particular record.

3.2 Source review of seven assigned studies

A subsequent AI-assisted source review under Raazia's direction checked the accessed full text of S008–S014, their 70 appraisal items and 13 reported outcomes. It corrected Clockwork PDF anchors and added its reported TVM v0.7 environment detail, separated Torpor's evaluated sharing mode from its runtime-isolation design and deployment, and clarified the two-stage request-rate setup of KRYPTON's QoS experiment. All 13 retained numerical values were found in the accessed sources; the changes concern interpretation, source anchors and appraisal rationale. The original forms are preserved, and the active workbook records original and reviewed QA scores separately.

Under the submitted claim-specific rubric, Clockwork's and StreamBox's Q4 and Q9 scores each rise from 1 to 2: hardware/software/workload information and explicitly delimited performance/trust boundaries are reported. The seven-study QA sum changes from 110/140 to 114/140. This is a reporting appraisal, not a new performance result or an independent human vote. ParvaGPU's numerical checks remain specific to the accessed arXiv v1 manuscript. The remainder of the corpus is outside this seven-study source-review pass.

3.3 Databases, search fields and period

The active occurrence register represents ACM Digital Library, IEEE Xplore, Springer Nature Link and ScienceDirect, with supplementary named-paper lookups recorded separately. Their relevance is complementary: systems conferences, parallel/distributed systems research and journal literature provide different routes to sharing mechanisms. However, representation in an imported register is not equivalent to verified complete coverage of every planned database run.

The main publication window is 2020 through the date of each executed search in October 2026. IEEE and Springer entries record 11 October; ACM and ScienceDirect entries record 14 October. Later retrieval of a paper does not change an earlier search cutoff. The older journal survey used as an organizational benchmark is contextual literature, not an additional included primary study.

The executed expressions generally combine GPU terms, inference/model-serving terms and mechanism terms. Some interfaces use broad indexed fields rather than title/abstract/keywords. Broad fields can retrieve papers in which sharing terminology occurs only in references or background. Conversely, restricted inference expressions can miss papers described only as LLM serving. The exact retained expressions, fields and filters are reproduced in Appendix A; the workbook holds filenames and provenance.

The historical ACM master record contains displayed, claimed and parsed totals of 1,600, 1,603 and 1,829. The ScienceDirect MPS record retains 517 displayed against 563 registered. These differences are not corrected by inventing a missing export or changing a count to match another artifact. The active consolidated occurrence totals and their original discrepancies are different fields.

3.4 Eligibility and study identity

All inclusion criteria must be met: an English peer-reviewed primary report; evaluated concurrent sharing on one physical GPU; inference outcomes, separable where training is

also present; interpretable outcomes and configuration; sufficient local technical evidence; and the specified publication period or justified older exception. A paper's title or prestigious venue alone does not establish eligibility.

Decisive exclusion categories are non-primary/non-peer-reviewed material, no evaluable inference-sharing mechanism, hardware-only work without evaluated systems sharing, aggregate-only multi-GPU evidence, non-GPU scope, insufficient assessed technical detail, and period/language mismatch. The supplied source assessment excludes TGS and HiveD because the extracted evaluation concerns training rather than eligible inference-sharing outcomes. These exclusions are kept outside the included-paper catalogue.

Records, reports and studies are distinct entities. A repeated database occurrence is linked during deduplication. A report version remains identifiable even when it belongs to the same underlying study as another report. REEF's conference report P005 and journal extension P023 map to S005; the journal's first page explicitly states the extension relationship [6, 7]. They cannot supply two independent replications. Other manuscript copies retain their version labels rather than automatically inheriting final-publication page numbers.

3.5 Appraisal and extraction

The Q1–Q10 rubric scores 0, 1 or 2 for objective clarity, mechanism description, experimental transparency, hardware/workload description, baseline fairness, utilization/throughput evidence, latency measurement, multi-tenant interference, isolation evaluation and replication/limitations. Justified not-applicable items reduce the applicable denominator. Scores are reporting and internal-validity appraisals, not journal-prestige scores or automatic exclusion criteria.

There are 200 appraisal items for 20 study-level forms. Three items are N/A in the supplied data: P001 Q7, P006 Q9 and P015 Q9. The normalized score is the recorded sum divided by the applicable maximum. The workbook recalculates these sums from the item rows and checks them against the study table. A companion report does not create an additional independent study-level appraisal.

Extraction preserves mechanism, environment, workload, tenants, load/SLO, baseline, result, isolation dimensions, overhead, limitations, source and anchor. The 52 outcome rows and two calculations are not 54 studies. Missing values remain missing or explicitly not established. A throughput ratio is not converted into utilization percentage, a model-pair distribution is not treated as a request-latency percentile, and a reported maximum is not recast as an average.

3.6 Synthesis and robustness rules

The synthesis is narrative and comparative. A numerical contrast is interpreted within the source experiment unless GPU, model, batch, load, concurrency, metric and baseline are sufficiently aligned across studies. No pooled speedup or weighted cross-study average is calculated. A result is first evaluated for relevance to the decision, then for comparability and appraisal limitations.

The protocol's below-60% sensitivity view removes no current study because all recorded normalized scores are at least 70%. Consequently, an unchanged answer under this cutoff is

mechanically expected. It does not demonstrate robustness to publication bias, title-only screening, a different stopping rule or correlated single-reviewer error. Claim-level anchors and version checks remain necessary even for high-scoring studies.

4 PRISMA study selection and corpus

The supplied logs reconcile arithmetically as 4,189 occurrences minus 114 duplicate links equals 4,075 canonical records. Their record-decision partition is 3,493 exclusions, 559 retained but not advanced, 21 included report versions and two source-assessed exclusions. Thus 582 records remain after screening, while only 23 report versions enter the bounded source-assessment set.

The source-assessment partition is 23 reports equals 21 included plus two excluded. The included reports map to 20 studies. The supplied historical flow records zero reports not retrieved within that selected assessment set. This is a historical statement in the baseline, not a claim that all PDFs are downloadable in the current environment. Task 3 retrieval status is separately logged and must not be used to overwrite historical PRISMA events.

The 559-record branch is especially consequential. These records are not technical exclusions and are not failed full-text retrievals: their reports were not sought under the stopping rule. Moving them into either category would misrepresent the actual method. The bounded scope requires explicit acceptance before describing the work as the final CEP review.

Of the 4,075 canonical records, 2,933 lack abstract text in the export and 1,142 have it. The recorded decisions for the former rely on title/type metadata. This limits confidence in screening recall even though every record has a decision field. A complete spreadsheet row does not guarantee a complete eligibility assessment.

Stage	**Count**
Records identified	4189
Duplicate links applied	114
Canonical records screened	4075
Excluded at record screening	3493
Retained after screening	582
Not advanced under stopping rule	559
Report versions assessed	23
Excluded after source assessment	2
Included report versions	21
Included unique studies	20

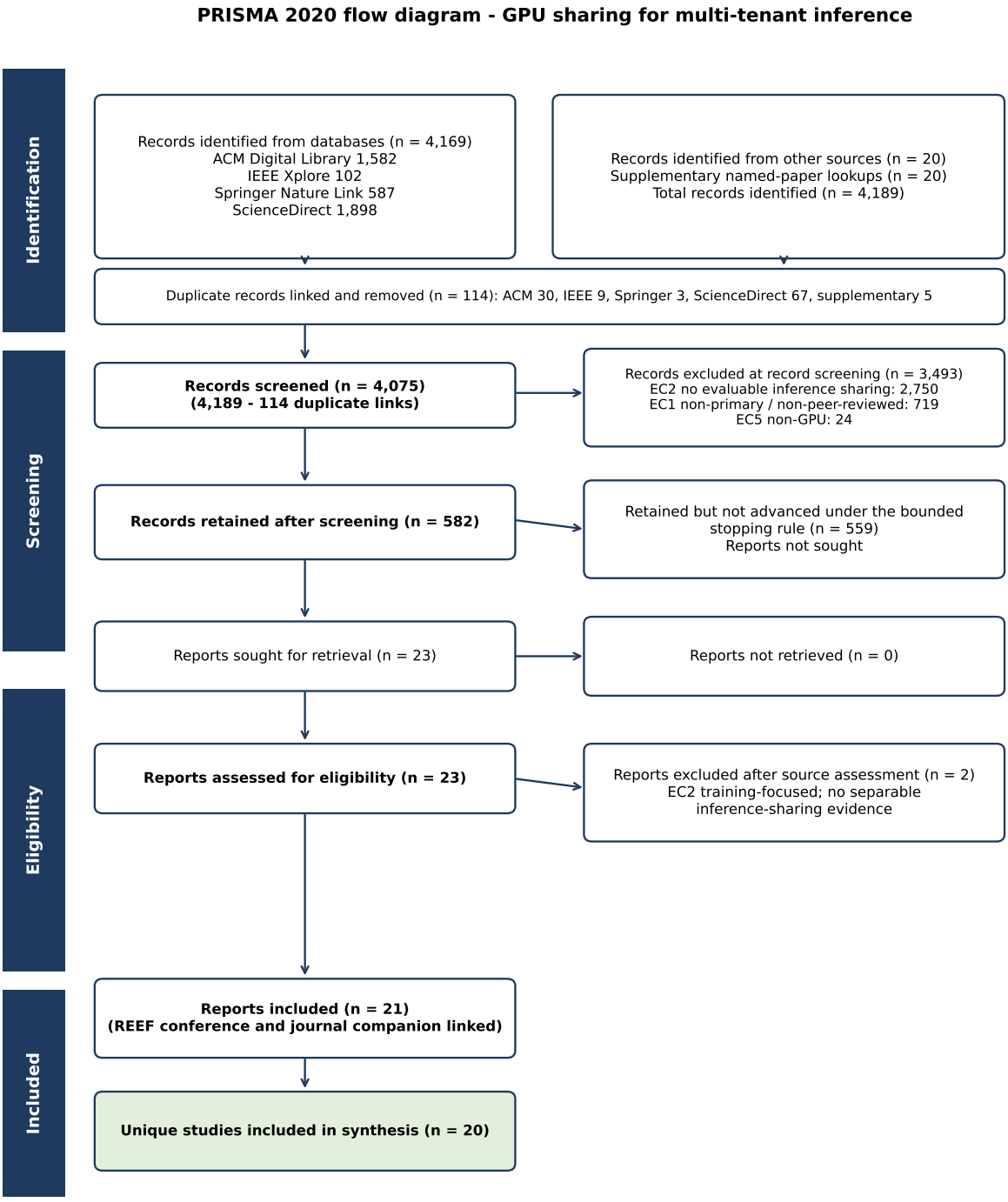


Figure 1: Supplied bounded-review selection flow, preserved from Task 2. Current PDF download attempts are a separate activity.

Study / report	System / source identity	QA
S001 / P001	VastPipe (2025)	13/18
S002 / P002	iGniter (2023)	17/20
S003 / P003	gpulets (2022)	18/20
S004 / P004	PipeSwitch (2020)	17/20
S005 / P005	REEF conference (2022)	17/20
S006 / P006	Paella: Low-latency Model Serving with Software-defined GPU Scheduling (2023)	16/18
S007 / P007	Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications (2024)	19/20
S008 / P008	GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform (2020)	17/20
S009 / P009	Serving DNNs like Clockwork: Performance Predictability from the Bottom Up (2020)	18/20
S010 / P010	Colocating ML Inference and Training with Fast GPU Memory Handover (SIRIUS) (2025)	17/20
S011 / P011	ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments (2024)	14/20
S012 / P012	Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference (2025)	15/20
S013 / P013	StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow (2024)	17/20
S014 / P014	Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception (KRYPTON) (2025)	16/20
S015 / P015	XSched: Preemptive Scheduling for Diverse XPUs (2025)	16/18
S016 / P018	InferFair: Towards QoS-aware scheduling for performance isolation guarantee in heterogeneous model serving systems (2024)	15/20
S017 / P019	ArkGPU: enabling applications' high-goodput co-location execution on multitasking GPUs (2023)	16/20
S018 / P020	Dyna-P: placement-aware dynamic partitioning for lightweight applications with modern GPUs (2025)	15/20

Study / report	System / source identity	QA
S019 / P021	KRISP: Enabling Kernel-wise RIght-sizing for Spatial Partitioned GPU Inference Servers (2023)	16/20
S020 / P022	Tally: Non-Intrusive Performance Isolation for Concurrent Deep Learning Workloads (2025)	17/20

5 Taxonomy of existing approaches

A useful taxonomy separates the resource being controlled from the software layer implementing the policy. The same system may appear in more than one category. A MIG-based system can also use MPS; an API interceptor can enforce temporal scheduling; and memory handover can support mixed-workload co-location. Treating these as mutually exclusive labels would hide the source of the observed behavior.

5.1 Temporal and priority control

Temporal mechanisms decide which work runs and when another tenant can make progress. PipeSwitch emphasizes switching and model state, Clockwork emphasizes predictable execution and admission, while REEF, Paella, Orion and XSched expose different scheduling/preemption boundaries. Their integration costs and latency meanings differ. PipeSwitch's additional switching-related latency is not interchangeable with XSched's foreground P99 ratio or Paella's reported P99 reduction [8, 6, 9, 10, 11, 2].

5.2 Spatial allocation and partitions

MPS-based allocation and software right-sizing regulate concurrent access to execution resources. VastPipe, iGniter, gpulets and GSLICE provide different allocation, profiling and feedback approaches. KRISP contributes AMD-oriented kernel right-sizing evidence and should not be relabeled as a measurement of NVIDIA MPS. Partition-aware designs, including ParvaGPU and Dyna-P, additionally make placement, profile size or reconfiguration relevant [12, 13, 14, 15, 1, 16, 17].

A spatial allocation is only one part of a service guarantee. The model must fit memory, the workload must remain within the profiled regime and the selected resource shares must satisfy its timing requirements. Hardware allocation boundaries do not by themselves establish all possible security or fault guarantees in the reviewed experiments.

5.3 API, runtime and memory mediation

Torpor, StreamBox, KRYPTON, ArkGPU and Tally intervene at different points between the application and accelerator. Their common advantage is control over resource use without necessarily changing the model's high-level purpose. Their differences include

intercepted interfaces, compatibility, runtime sharing, host work and the degree to which tenants share memory or trust assumptions. SIRIUS contributes a specific memory-handover mechanism for co-located inference and training [18, 19, 20, 21, 22, 23].

Family / reports	**Control and condition**	**Cost**
Temporal / priority; P004 P005 P006 P007 P009 P015 P022	Request, model, kernel or launch order. A suitable interception/preemption boundary exists	Switch cost, profiling and background slowdown
Software spatial sharing; P001 P002 P003 P008 P021	MPS quota or AMD CU-mask control. Concurrent jobs fit memory and selected compute allocation	Residual interference; quotas are not proof of security
Hardware partition and hybrid; P011 P014 P020	MIG profiles plus temporal/MPS policy. Supported partition profiles fit model memory	Fragmentation, placement and reconfiguration cost
API / runtime mediation; P006 P012 P013 P014 P019 P022	User API, shared runtime or kernel interface. Required APIs and integration hooks are supported	Compatibility, CPU/transfer cost and trust boundary
Memory handover; P010	Inference/training memory ownership. Training can yield memory without breaking inference	Reclamation and integration overhead
Fairness policy; P018	Relative model slowdown and SLO goodput. Comparable per-model service objectives	Not equivalent to request-tail or hostile-tenant isolation

6 Results and evidence synthesis by research question

6.1 RQ1: mechanisms and control boundaries

The corpus supports a layered answer rather than a product list. Temporal control manages ordering and interruption; software spatial control manages concurrent allocation; hardware partitioning constrains resource placement; and API/runtime mediation supplies an enforcement layer. Fairness-aware policy adds a decision objective across these mechanisms. InferFair, for example, is best understood through relative slowdown and useful service, not as a new hardware partition primitive [24].

This taxonomy explains why naming MPS, MIG or remoting is insufficient for engineering selection. A deployment also needs a policy, a mechanism for enforcing it and an explicit boundary of protection. The corpus demonstrates several combinations, but its bounded

selection does not support a claim that one category is most prevalent in the entire literature.

6.2 RQ2: measurements and comparability

The outcome register includes requests/s, normalized throughput, goodput, P99/P98 latency, unspecified latency, SLO-violation rate, API overhead, host memory and a composite utilization-efficacy score. These measure different properties. Goodput counts work that meets a service objective, while aggregate throughput may include best-effort work that does not protect the foreground service. A numerical increase in the latter cannot establish improvement in the former.

GSLICE's GUE value of 94.04 is a composite score, not 94.04% GPU utilization. Its negative signed latency improvement is retained as an unfavorable observation. The gpulets threshold covering 90% of tested model pairs concerns a distribution across pairs, not the 90th percentile of requests. These distinctions prevent apparently precise but invalid conclusions [14, 15].

The reported iGniter prediction-error range of 1.53–5.02% assesses prediction accuracy, not a latency speedup. Clockwork's approximately 800 requests/s is goodput in a particular evaluation, and the supplied extraction notes a nonidentical comparator setup. Both observations are useful, but neither supports a direct ranking against a throughput ratio measured on another platform [13, 11].

6.3 RQ3: conditions affecting results

Three conditions recur in the evidence. First, concurrency and resource shares affect both completed work and waiting time. The local ParvaGPU contrast shows a throughput benefit accompanied by increased latency. Second, model state and memory transfer can dominate. Torpor's late binding is attractive for low-rate functions, but its density result does not establish that frequent transfers of large models meet the same latency objective. Third, the workload's priority structure determines which sacrifice is acceptable [1, 19].

For a foreground-priority service, XSched's 1.07x standalone P99 is relevant because it directly expresses foreground interference in the evaluated setup. Orion's up-to-7.3x aggregate-throughput result describes total work and must be read with its foreground-latency condition. These results can support different decisions without contradicting each other [10, 2].

Mixed-workload evidence requires an additional separation. SIRIUS reports an average P99 inference improvement factor of 12.6 against the authors' adapted TaskSwitch baseline. It is not a uniform per-model gain or a direct comparison against unmodified PipeSwitch. ArkGPU's recorded inference/training pairs meet a twice-solo P99 target, while a low-contention setting can lose about 5% QPS. The unfavorable condition belongs in the synthesis because it changes when the mechanism is worthwhile [18, 22].

6.4 RQ4: isolation and implementation cost

Performance isolation asks whether another workload changes service behavior. Resource isolation asks what capacity is allocated or separated. Memory protection and fault

containment concern different failure or adversarial models. A scheduler that limits slowdown is not thereby a complete hostile-tenant security boundary. The extraction cards preserve these dimensions separately, and untested boundaries remain unestablished.

Implementation costs also operate at different levels. API mediation can add host-side work or per-call delay. Model swapping adds transfer and memory pressure. Kernel-space interception introduces compatibility and maintenance requirements. Temporal granularity changes the trade-off between scheduling work and waiting time. The KRYPTON extraction reports an average QoS violation rate of 1.23% relative to a solo P99 threshold, which is a threshold-relative request fraction rather than a latency in milliseconds. Its original Poisson rate is half the solo maximum; the shared test halves that rate again while assigning 50% time [21].

6.5 RQ5: conditional selection

The evidence supports a sequence of decisions. Specify the service objective and trust boundary first. Eliminate mechanisms incompatible with hardware, model memory or required integration. Compare the remaining candidates under the relevant arrival process, batch and co-tenant mix. Measure both protected service and work sacrificed. If the available studies do not cover the required boundary, the appropriate conclusion is insufficient evidence for that guarantee, not an assumed benefit.

The recommendation matrix in Section 10 connects each scenario to report IDs, expected benefits, costs and alternatives. These are evidence-supported candidates for evaluation, not claims that the reviewed implementation can be deployed unchanged in every environment.

7 Comparative analysis

The comparison distinguishes direct within-source contrasts from unmatched cross-study observations. The strongest numerical statements are tied to their original comparator. Even these require attention to whether the result is an average, maximum, lower bound or threshold-relative statistic. An improvement described as “up to” is not the expected result for an arbitrary deployment.

Evidence / class	Observation	Interpretation and missing control
CMP01 P011 E036 E037 C001 C002. Within-source trade-off	354 to 446 requests/s; 11 to 27 ms	25.99% throughput increase accompanies 2.4545x latency. Missing: Latency quantile; final version equivalence
CMP02 P001 E001; P011 E036. Unmatched cross-study	2.7x versus 25.99% increase	Different baseline and GPU/model context prevent ranking. Missing: Matched workload, batch and resource budget

Evidence / class	Observation	Interpretation and missing control
CMP03 P007 E029 E030; P015 E044. Different objectives	Up to 7.3x aggregate work versus 1.07x foreground P99	Foreground delay and total completed work must be assessed together. Missing: Common model mix and background progress metric
CMP04 P008 E031 E032. Within-source metric tension	GUE 94.04 and negative signed latency improvement	Composite efficacy is not utilization percent; preserve adverse latency. Missing: Metric definitions and baseline stratum
CMP05 P012 E038 E039. Condition-dependent trade-off	Density benefit with model-transfer sensitivity	Late binding suits low-rate functions only if transfer fits SLO. Missing: Heavy-model transfer and host CPU budget
CMP06 P019 E048 E049; P022 E052. Different experiments, consistent cost	QoS protection with low-contention or background throughput loss	Protecting foreground work can sacrifice co-tenant work. Missing: Matched slowdown definitions
CMP07 P005 P023. Companion study, not replication	REEF engines and report versions differ	Retain source-version strata; do not double-count corroboration. Missing: Outcome-to-version mapping for every combined REEF row
CMP08 P018 E046 E047; P020 E050. Different metrics	Goodput/fairness versus makespan	Job-level efficiency cannot establish request-tail service. Missing: Request arrival process and latency distribution

The evidence table in Appendix C retains all 52 supplied observations. The comparative matrix is deliberately smaller because its role is to explain a decision, not duplicate the extraction register. A claim can be traced from a recommendation to a report, then to an outcome ID and its page/figure anchor. Where a comparison remains unmatched, the matrix names the missing control rather than calculating a misleading combined effect.

8 Engineering trade-off analysis

8.1 Utilization versus latency

Additional concurrency is beneficial only while useful service increases under the required latency objective. The ParvaGPU example changes throughput by

$$(446/354 - 1) \times 100 = 25.99\%,$$

while latency changes by

$$27/11 = 2.4545.$$

These are calculations from recorded inputs, not newly measured outcomes. A service with a tight delay budget might reject this configuration despite its higher throughput. Another service might accept it if the increased latency remains within its own budget. Because the source quantile is unspecified, neither interpretation establishes P99 compliance [1].

8.2 Foreground protection versus background progress

Priority is an allocation choice. Tally's supplied contrast records normalized background throughput of 1.19 versus 1.65 for TGS at the stated idle-time setting. This does not make foreground protection ineffective; it identifies the work sacrificed. Similarly, an aggregate-throughput gain must not hide deterioration in the protected request class. The engineering evaluation should report foreground latency and background progress together [23].

8.3 Memory density versus transfer and trust costs

Shared runtimes and late binding can allow more functions or models to use limited GPU memory. StreamBox's workflow-oriented sharing and Torpor's binding decisions address this problem from different directions. Their density gains cannot be separated from concurrency, model-loading and trust assumptions. When large models are invoked frequently, transfer demand can undermine the benefit of keeping fewer models resident. When functions require stronger separation, a shared runtime may not provide the required boundary. Torpor explicitly distinguishes runtime-sharing evaluation from runtime-isolation deployment, and describes restart/migration recovery; these design claims are not direct fault-injection results [19, 20].

8.4 Partition control versus flexibility

Hardware profiles constrain placement and memory fit. Software policies can improve use of a chosen profile, but fine-grained application demand need not align with fixed partitions. Dyna-P's recorded sensitivity to prediction perturbation concerns makespan, while online serving additionally needs evidence about individual request delays. The review therefore treats placement efficiency as relevant but insufficient evidence for latency-sensitive request serving [16].

9 Discussion and conflicting evidence

9.1 Apparent conflicts versus controlled disagreements

A valid conflicting-evidence analysis first asks whether two studies measure the same quantity under comparable conditions. VastPipe's 2.7x throughput contrast and the calculated 25.99% ParvaGPU increase have different baselines and workloads. Their unequal magnitudes do not show disagreement. Likewise, Orion's aggregate throughput, SIRIUS's average P99 improvement and XSched's normalized foreground P99 do not compete on a common scale [12, 10, 18, 1, 2].

More useful tensions occur inside an experiment: improved throughput accompanied by higher latency, or protection of foreground work accompanied by lower background progress. The negative GSLICE latency contrast and ArkGPU's low-contention regression show conditions in which a favorable overall mechanism narrative is incomplete. These observations constrain applicability rather than being dismissed as exceptions [15, 22].

9.2 Version and engine effects

The REEF journal extension adds implementation and evaluation beyond the conference report. The supplied extraction already distinguishes PyTorch and TensorRT conditions in its discussion. Combining all favorable points into one synthetic result would obscure both the engine difference and the report-version difference. The current study-level mapping avoids double-counting, but each combined REEF outcome still needs an explicit conference-versus-journal source assignment before final submission [6, 7].

9.3 Strength of the engineering argument

The repeated pattern is that resource sharing works through a mechanism-specific constraint: interruption cost, quota fit, memory residency, partition placement or compatibility. This pattern supports conditional design reasoning. It does not establish identical effect sizes or a common security guarantee across systems. The draft's recommendations remain narrower than a claim of deployment readiness.

10 Evidence-based engineering recommendations

REC01: Trusted resident models; utilization priority

Dimension	Assessment
Candidate	Tuned MPS / software spatial controls
Evidence	P001 P002 P003 P008 E001-E004 E006 E031-E032
Expected benefit	Higher useful co-location under tested allocations
Sacrifice	Interference, quota tuning and memory pressure

Dimension	Assessment
Alternative / trigger	Partitions or dedication when trust/isolation dominates
Validation needed	Measure each model P99 and SLO goodput under bursts

REC02: Foreground tail-latency priority

Dimension	Assessment
Candidate	Priority/preemptive scheduling
Evidence	P005 P006 P007 P015 P022 E018-E030 E044-E045 E052
Expected benefit	Lower foreground interference
Sacrifice	Background slowdown, profiling and runtime integration
Alternative / trigger	Dedicated GPU when supported control cannot meet deadline
Validation needed	Check foreground tails and background progress jointly

REC03: Low-rate serverless functions

Dimension	Assessment
Candidate	Late binding / controlled runtime sharing
Evidence	P012 P013 E038-E041
Expected benefit	Higher function or workflow density
Sacrifice	Host CPU, memory transfer and shared-runtime trust
Alternative / trigger	Keep large or high-rate models resident if swaps dominate
Validation needed	Sweep model sizes, cold starts, offered rate and concurrency

REC04: Mixed inference with opportunistic training

Dimension	Assessment
Candidate	Memory handover / controlled co-location
Evidence	P010 P019 E035 E048-E049

Dimension	Assessment
Expected benefit	Protect inference while using residual capacity
Sacrifice	Reclamation overhead and reduced training progress
Alternative / trigger	Separate resources for intolerable interference
Validation needed	Measure inference tails separately from training throughput

REC05: Supported partition hardware; stable memory fit

Dimension	Assessment
Candidate	MIG-aware hybrid policy
Evidence	P011 P014 P020 E036-E037 E042-E043 E050
Expected benefit	Resource allocation with configurable co-location
Sacrifice	Fragmentation and reconfiguration overhead
Alternative / trigger	Software sharing when partition granularity wastes capacity
Validation needed	Validate profile fit and reconfiguration effects; security needs separate tests

REC06: Several models with unequal service pressure

Dimension	Assessment
Candidate	Fairness-aware admission / scheduling
Evidence	P018 E046-E047
Expected benefit	Balance normalized slowdown and SLO goodput
Sacrifice	May give up aggregate throughput to improve fairness
Alternative / trigger	Priority scheduling when one foreground SLA dominates
Validation needed	Agree fairness objective and per-model SLO before deployment

The recommended deployment assessment begins with the actual service objective: per-model deadline or percentile target, admissible violation rate and minimum background progress. These values must come from the intended deployment rather than being invented from a paper. The next step is a local comparison using the same model versions, input

sizes, batching and offered load. Cold starts and bursts should be retained if they occur in the application.

A candidate is suitable only if it meets the required service objective at an acceptable cost. The acceptance evidence should include per-tenant latency, SLO-compliant goodput, memory pressure, resource utilization with a defined instrument, host overhead and the relevant isolation tests. The review supplies a shortlist and the reasons to test it; it does not supply a universal numerical threshold or imply that a benchmark result transfers unchanged.

11 Research gaps and future directions

Corpus gap	**Basis**	**Next experiment**
No matched all-family comparison	Comparability CMP01-CMP08	Same GPU, model mix, memory budget, arrivals and SLO across families
Security/fault boundaries often unestablished	Extraction Security boundary / Fault boundary	Separate adversarial access, fault injection and performance tests
Metrics emphasize different units	E001-E052	Report per-model goodput, tails and background progress jointly
Partition prediction affects makespan	P020 E050	Bursty arrivals plus placement changes and request-tail measurement
Version and method details remain incomplete	P002 P011 P022; QA Q3/Q4/Q10	Publish exact version, driver/runtime, warmup, repetitions and uncertainty
Energy and cost evidence not harmonized	Extraction schema and heterogeneous outcomes	Measure GPU and host energy per SLO-compliant request

These gaps are observations about this corpus and its reporting, not assertions that no relevant research exists anywhere. The 559-record unassessed branch makes that distinction essential. A stronger future comparison would align hardware, workload, baselines and outcome definitions across mechanism families, then report both benefits and sacrifices. Energy per useful request and failure/security behavior require separate measurements rather than extrapolation from utilization.

12 Threats to validity and limitations

Threat / consequence	Mitigation	Remaining work
Selection/stopping: No exhaustive recall or prevalence claim	Stopping branch remains visible	Instructor/team decision on bounded scope; extend if required
Title-only decisions: Relevant work may be missed	Abstract-availability column retained	Prioritize uncertain/relevant records for abstract checks
Single review: Correlated screening/QA/extraction errors possible	Original rationale and source anchors retained	Record real named checks if performed
Search provenance: Search cannot be fully reconstructed from totals	Exact preserved query/log values retained	Supply missing original exports, facets and receipts
Source version: Page/result mappings may differ	Separate source-version and retrieval records	Compare final published text and map companion outcomes
Measurement heterogeneity: Cross-study speedup ranking is invalid	No pooled effect size or universal winner	Matched comparisons needed
Reporting bias: Headlines may overstate robustness	Adverse and null findings retained	Recheck high-impact claims at exact source locations

The most material limitation is selection: a bounded source set cannot establish exhaustive coverage. The large number of title-only records compounds that risk. The single-assessment method creates a separate risk of correlated errors even where spreadsheet arithmetic is correct. Finally, retrieved author manuscripts may differ from the version used by the original extraction, and access to a PDF does not constitute verification of every result.

Mitigations in this draft include immutable baseline files, separate current retrieval records, stable IDs, preserved unfavorable findings, explicit source/version limits and no numerical pooling across unmatched studies. These measures improve traceability but do not erase the underlying limitations. Final acceptance requires resolving the scope decision and recording any actual independent checks at the item level.

13 Conclusion

The bounded 20-study corpus supports a conditional approach to GPU sharing for multi-tenant inference. Temporal control is relevant when foreground response dominates; software spatial sharing is relevant when trusted co-location can exploit spare capacity; partition-aware designs make resource fit and reconfiguration central; and runtime or memory mediation can increase density at a transfer, integration or trust cost. Each choice must be evaluated against its own baseline and intended conditions.

The principal outcome is therefore a traceable selection framework, not a universal ranking. The first draft covers the instructor's analytical requirements through Section 22 and follows the suggested final-report sequence. It remains a draft: the bounded scope, source-version mappings, incomplete PDF collection and item-level checking require explicit closure before the work is represented as a fully verified final SLR.

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A Exact search register

The following entries are preserved from the supplied workbook. Date precision, provenance and count discrepancies remain as recorded.

A.1 IEEE-MPS

Field	Recorded value
Platform	IEEE Xplore
Status	executed search record preserved in the project register
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND (MPS OR "multi-process service")
Fields	All metadata (Search by All); not field-qualified Command Search
Filters	Publication years2020-2026; date/language/type checked during screening
Recorded search date / time	2026-10-11
Displayed hits	25
Exported records	25
Parsed records	25
Active records	25
Files	`https://github.com/Raazia-Imran/os_CEPPPPPP/blob/main/evidence/imports/group-a/imported_records.json`
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Group A run, reused (counts identical to Group A report, 16 Oct 2026)

A.2 IEEE-MIG

Field	Recorded value
Platform	IEEE Xplore

Field	Recorded value
Status	executed search record preserved in the project register
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND (MIG OR "multi-instance GPU" OR "GPU partitioning")
Fields	All metadata (Search by All); not field-qualified Command Search
Filters	Publication years2020-2026; date/language/type checked during screening
Recorded search date / time	2026-10-11
Displayed hits	23
Exported records	23
Parsed records	23
Active records	23
Files	`https://github.com/Raazia-Imran/os_CEPPPPPP/blob/main/evidence/imports/group-a/imported_records.json`
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Group A run, reused (counts identical to Group A report, 16 Oct 2026)

A.3 IEEE-TEMPORAL

Field	Recorded value
Platform	IEEE Xplore
Status	executed search record preserved in the project register
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching")
Fields	All metadata (Search by All); not field-qualified Command Search

Field	Recorded value
Filters	Publication years2020-2026; date/language/type checked during screening
Recorded search date / time	2026-10-11
Displayed hits	45
Exported records	45
Parsed records	45
Active records	45
Files	`https://github.com/Raazia-Imran/os_CEPPPPPP/blob/main/evidence/imports/group-a/imported_records.json`
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Group A run, reused (counts identical to Group A report, 16 Oct 2026)

A.4 IEEE-API

Field	Recorded value
Platform	IEEE Xplore
Status	executed search record preserved in the project register
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
Fields	All metadata (Search by All); not field-qualified Command Search
Filters	Publication years2020-2026; date/language/type checked during screening
Recorded search date / time	2026-10-11
Displayed hits	9
Exported records	9

Field	Recorded value
Parsed records	9
Active records	9
Files	`https://github.com/Raazia-Imran/os_CEPPPPPP/blob/main/evidence/imports/group-a/imported_records.json`
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Group A run, reused (counts identical to Group A report, 16 Oct 2026)

A.5 SPR-MASTER

Field	Recorded value
Platform	Springer Nature Link
Status	executed search record preserved in the project register
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching" OR MPS OR "multi-process service" OR MIG OR "multi-instance GPU" OR "GPU partitioning" OR "API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
Fields	General search/full-text indexed collection
Filters	English; Research article OR Conference paper;2020-2026; manual cutoff
Recorded search date / time	2026-10-11
Displayed hits	587
Exported records	587
Parsed records	587
Active records	587
Files	`https://github.com/Raazia-Imran/os_CEPPPPPP/blob/main/evidence/imports/group-a/imported_records.json`

Field	Recorded value
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Group A run, reused (counts identical to Group A report, 16 Oct 2026)

A.6 ACM-PILOT

Field	Recorded value
Platform	ACM Digital Library
Status	Blocked pilot; no records imported
Exact query	GPU AND inference AND MPS
Fields	not_validated
Filters	not_applied
Recorded search date / time	2026-10-11
Displayed hits	UNKNOWN
Exported records	0
Parsed records	0
Active records	0
Files	supplied search `export/imported_records.json`
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Pilot checkpoint of 11 Oct 2026 (search blocked/unavailable; no import), as in Group A report

A.7 SD-PILOT

Field	Recorded value
Platform	ScienceDirect
Status	Blocked pilot; no records imported

Field	Recorded value
Exact query	GPU AND inference AND MPS
Fields	not_validated
Filters	not_applied
Recorded search date / time	2026-10-11
Displayed hits	UNKNOWN
Exported records	0
Parsed records	0
Active records	0
Files	supplied search `export/imported_records.json`
Notes	Rendered metadata capture, not publisher export; exact time unavailable; different expressions from submitted protocol; full text indexed scope for Springer
Provenance	Pilot checkpoint of 11 Oct 2026 (search blocked/unavailable; no import), as in Group A report

A.8 SUP-SELECTED

Field	Recorded value
Platform	Supplementary public web/venue/author lookup
Status	executed search record preserved in the project register
Exact query	Named-paper lookups and official venue/author retrieval; see `supplementary_discovery.csv`
Fields	Public metadata/full texts
Filters	Same protocol eligibility
Recorded search date / time	2026-10-11
Displayed hits	UNKNOWN
Exported records	20
Parsed records	20
Active records	20

Field	Recorded value
Files	`evidence/imports/group-a/supplementary_discovery.csv`
Notes	20 selected named-paper records in the preserved source register; this is not an exhaustive web search. Query wording is a reproducible follow-up, not a verbatim historical query.
Provenance	Shared original-collector register, retained with its source IDs; no new search claimed.

A.9 ACM-MASTER

Field	Recorded value
Platform	ACM Digital Library
Status	reported master query; master and numbered exports not fully mapped
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning" OR DNN OR LLM) AND ("time slicing" OR "time sharing" OR "context switching" OR MPS OR "multi-process service" OR MIG OR "multi-instance GPU" OR "GPU partitioning" OR "API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
Fields	All Fields
Filters	2020 - 2026
Recorded search date / time	2026-10-14 22:30 PKT (UTC+05:00; operator-reported)
Displayed hits	1600
Exported records	1603
Parsed records	1829
Active records	0
Files	`evidence/incoming_shared/acm_master/ACM_MASTER_01.bib;evidence/incoming_shared/acm_master/ACM_MASTER_02.bib`

Field	Recorded value
Notes	Separate master exports contain 1,829 overlapping occurrences / 1,387 distinct DOI-or-title keys. Reported displayed 1,600 and claimed export 1,603 do not reproduce those bytes. Active contribution is zero; do not assign the 1,582 numbered occurrences to this run without mapping evidence. Source: `evidence/incoming_shared/acm_master/reconciliation.json`.
Provenance	Historical operator-reported run; timestamp/query from `evidence/incoming_shared/search_log_2026-10-16.txt`; not independently rerun.

A.10 SD-TEMPORAL

Field	Recorded value
Platform	ScienceDirect
Status	historical operator-reported; partial export reconciliation
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND ("time slicing" OR "time sharing" OR "context switching")
Fields	Find articles with these terms
Filters	2020-2026; supplied URL explicitly includes accessTypes=openaccess (not stated in original filter text).
Recorded search date / time	2026-10-14 23:45 PKT (UTC+05:00; operator-reported)
Displayed hits	1053
Exported records	1053
Parsed records	1053
Active records	1053

Field	Recorded value
Files	`evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_01.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_010.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_011.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_02.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_03.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_04.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_05.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_06.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_07.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_08.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_TEMPORAL_09.ris`
Notes	TEMPORAL URL has openaccess facet omitted from filter text; 1053 parsed agrees with reported count; exact pagination and operator unknown Repository check: standalone `evidence/incoming_shared/SD_TEMPORAL_01(1).ris` contains 25 records only; it is not the 1053-occurrence full TEMPORAL archive cited in the project register. Later archive files are not present in the inspected remote tree. Original reported/registered counts are preserved, not revalidated from this sample.
Provenance	Historical operator-reported run; timestamp/query from `evidence/incoming_shared/search_log_2026-10-16.txt`; not independently rerun.

A.11 SD-MPS

Field	Recorded value
Platform	ScienceDirect
Status	historical operator-reported; partial export reconciliation
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND (MPS OR "multi-process service")
Fields	Find articles with these terms
Filters	2020 - 2026

Field	Recorded value
Recorded search date / time	2026-10-14 23:52 PKT (UTC+05:00; operator-reported)
Displayed hits	517
Exported records	563
Parsed records	563
Active records	563
Files	`evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_01.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_02.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_03.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_04.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MPS_05.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SSD_MPS_06.ris`
Notes	563 parsed versus 517 displayed; export claim 563; operator and exact pagination unknown Repository check: standalone `evidence/incoming_shared/SD_MPS_01.ris` contains 25 records only; it is not the 563-occurrence full MPS archive cited in the project register. Later archive files are not present in the inspected remote tree. Original reported/registered counts are preserved, not revalidated from this sample.
Provenance	Historical operator-reported run; timestamp/query from `evidence/incoming_shared/search_log_2026-10-16.txt`; not independently rerun.

A.12 SD-MIG

Field	Recorded value
Platform	ScienceDirect
Status	historical operator-reported; partial export reconciliation
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND (MIG OR "multi-instance GPU" OR "GPU partitioning")
Fields	Find articles with these terms
Filters	2020 - 2026

Field	Recorded value
Recorded search date / time	2026-10-14 23:55 PKT (UTC+05:00; operator-reported)
Displayed hits	241
Exported records	241
Parsed records	241
Active records	241
Files	`evidence/incoming_shared/sciencedirect_2026-10-16/SD_MIG_01.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MIG_02.ris;evidence/incoming_shared/sciencedirect_2026-10-16/SD_MIG_03.ris`
Notes	Parsed count agrees with reported count; identity and page completeness unverified; operator unknown Repository check: standalone `evidence/incoming_shared/SD_MIG_01.ris` contains 25 records only; it is not the 241-occurrence full MIG archive cited in the project register. Later archive files are not present in the inspected remote tree. Original reported/registered counts are preserved, not revalidated from this sample.
Provenance	Historical operator-reported run; timestamp/query from `evidence/incoming_shared/search_log_2026-10-16.txt`; not independently rerun.

A.13 SD-API

Field	Recorded value
Platform	ScienceDirect
Status	historical operator-reported; partial export reconciliation
Exact query	(GPU OR "graphics processing unit") AND (inference OR "model serving" OR "deep learning") AND ("API remoting" OR "CUDA remoting" OR "GPU proxy" OR "GPU virtualization")
Fields	Find articles with these terms
Filters	2020 - 2026
Recorded search date / time	2026-10-14 23:56 PKT (UTC+05:00; operator-reported)

Field	Recorded value
Displayed hits	41
Exported records	41
Parsed records	41
Active records	41
Files	`evidence/incoming_shared/sciencedirect_2026-10-16/SD_API_01.ris`
Notes	Parsed count agrees with reported count; identity and page completeness unverified; operator unknown Repository check: standalone `evidence/incoming_shared/SD_API_01.ris` contains 25 records only; it is not the 41-occurrence full API archive cited in the project register. Later archive files are not present in the inspected remote tree. Original reported/registered counts are preserved, not revalidated from this sample.
Provenance	Historical operator-reported run; timestamp/query from `evidence/incoming_shared/search_log_2026-10-16.txt`; not independently rerun.

A.14 ACM-NUMBERED-UNMAPPED

Field	Recorded value
Platform	ACM Digital Library
Status	actual supplied exports; run mapping unknown
Exact query	unknown
Fields	unknown
Filters	unknown
Recorded search date / time	unknown
Displayed hits	unknown
Exported records	unknown
Parsed records	1582
Active records	1582

Field	Recorded value
Files	`evidence/imports/acm/*.bib`; exact files in import_manifest.csv
Notes	79 preserved files contain 1,582 occurrences with hashes in `evidence/import_manifest.csv`. Filename number 3 is absent; this is not proof of a missing page. Exact query/date/field mapping for these files is not documented. Keep separate from the two overlapping master exports.
Provenance	Original user-supplied numbered ACM exports; repository manifest supports count and hashes, not historical run mapping.

B Complete study appraisal and extraction cards

These cards use the supplied study forms with the documented AI source-review overlay for S008–S014. Original forms remain in the frozen baseline. They are an audit appendix; the main text is organized by mechanism and RQ. Scores are not a claim of independent verification.

B.1 S001 / P001 VastPipe

[12]

Field	Extracted content
Mechanism	MPS adaptive quotas; separate AMD CU-mask stratum
Hardware and environment	NVIDIA A100 for MPS; AMD MI100 for CU-mask tests
Runtime	NVIDIA TensorRT serving; AMD TVM/ROCm with `hipExtStreamCreateWithCUMask`
Workload	CV pairs and BERT/GPT2 mixed model pairs; Poisson-generated request mixes
Tenants	Two co-located inference model services in reported local model-pair comparisons
Load and SLO	Poisson-generated mixed requests in batch-oriented co-location tests. Specific GPU-test arrival rates and latency SLO are not established in the current extract.
Comparator	No MPS; default MPS without per-service quotas; RL-selected MPS quotas. AMD no/default CU mask separately.
Finding	A100 VGG16+GoogLeNet: throughput 2.7x no MPS and 1.5x default MPS; not a universal average or p99 result. NLP/CV mean 2.43x no MPS, 1.45x default MPS. MI100 mean 1.89x no mask, 1.24x default mask.
Performance isolation	Throughput/interference comparison only; no extracted tail-latency or SLO protection measurement.
Security boundary	Not established
Fault boundary	Not tested in extracted GPU results
Overhead	Offline Q-table generation; plans reused within request windows; source Section 6.9
Limitations	Include only local GPU inference evidence, not the main 1000-NPU cluster results. Offline Q-table assumes relatively stable request mix; GPU-specific quotas, repetitions and variance unresolved. Chart bars are not preserved in Markdown. Preliminary ICCD2021 report is linked as a companion-version lead outside the synthesis corpus.
Source	`https://doi.org/10.1007/s11390-024-3773-5`
Evidence location	Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Objective and local GPU portability question explicit. Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q2	2 / 2	Tuned MPS quotas and CU masking described. Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q3	1 / 2	GPU-specific repeats and variance unresolved in conversion. Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q4	1 / 2	GPU/backend named; precise GPU loads and versions incomplete. Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q5	1 / 2	Same hardware and no/default/tuned MPS; quotas incomplete. Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q6	2 / 2	Throughput ratios and baseline named in prose. Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q7	NA / 0	GPU-portability extraction makes no latency claim; do not score NPU latency as GPU evidence. Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q8	2 / 2	Multiple named co-located inference pairs evaluated. Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q9	1 / 2	Performance interference considered; no separate security/fault validation extracted. Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16
Q10	1 / 2	Stable-mix Q-table constraint; GPU replication settings incomplete. Published pp. 458-459; PDF pp. 16-17; Sections 6.7-6.9; Figures 14-16

B.2 S002 / P002 iGniter

[13]

Field	Extracted content
Mechanism	MPS active-thread quotas + interference-aware batch/provisioning and shadow process
Hardware and environment	Ten EC2 p3.2xlarge instances,one V100 each,8vCPU/61GB; `Triton2.12.0,TensorRT8.0.1.6.` MPS active-thread percentage; allocation unit2.5%. Heterogeneous extension adds20 T4 instances.
Runtime	NVIDIA Triton 2.12.0; TensorRT 8.0.1.6; MPS active-thread percentage (recorded hardware/setup evidence).
Workload	`AlexNet,ResNet50,VGG19(ImageNet),SSD(VOC2012);12` workload/SLO configurations; constant arrivals. Local experiment co-locates four models at25% quota,batch3. p99 provisioning measured over windows e.g.30s; model-validation experiments repeated3 times with SD bars.
Tenants	Local test co-locates four inference models at 25% MPS quota each, batch 3; separate provisioning experiments use multiple instances.
Load and SLO	batch3 for prediction check;W1 SLO10ms for recovery example
Comparator	FFD+; GSLICE+ with iGniter placement; gpu-lets+ modified batch strategy and best-fit placement. These are adapted baselines.
Finding	Local four-model latency prediction `error1.53%-5.02%(Fig13).` Fig14 deployment: iGniter6 GPUs/0 SLO violations vs `gpu-lets+8/3,FFD+5/10,GSLICE+6/3;25%` lower hourly instance cost vs gpu-lets+, not25% higher local utilization. Fig17 shows an initial p99 violation before shadow-process recovery.
Performance isolation	Local interference prediction and p99/SLO checks; transient prediction failure explicitly retained
Security boundary	Not established
Fault boundary	Not tested in extracted results
Overhead	Shadow standby consumes capacity/memory; extra provisioning and interference-model costs are discussed. A standalone numerical overhead is not assessed in the retained extract.

Field	Extracted content
Limitations	Constant arrivals and four vision models limit bursty/LLM transfer. Negligible PCIe contention assumption can underpredict AlexNet; baselines modified. `Steady-state/post-correction` SLO success is not an unconditional guarantee; shadow standby incurs capacity/memory costs. Chart bars/latency traces unavailable in Markdown.
Source	`https://doi.org/10.1109/TPDS.2022.3232715`
Evidence location	Published pp820-824; sections4.2,5.1-5.4; Figs13-18; Markdown lines938-1245

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Interference-aware inference provisioning objective explicit. abstract,section1
Q2	2 / 2	MPS allocation and shadow implementation described. section4.2
Q3	2 / 2	Model checks repeated three times with SD; metric windows specified. sections5.1-5.3
Q4	1 / 2	GPU/workloads/backend detailed; OS/driver unresolved. sections4.2,5.1
Q5	1 / 2	Adapted baselines and allocation/batch differences disclosed. section5.1
Q6	2 / 2	Cost, throughput, resource-allocation metrics distinguished. sections5.1,5.3
Q7	2 / 2	P99 windows, SLO and recovery timing explicit. section5.3 Fig17
Q8	2 / 2	Four co-located models and quota configuration specified. section5.2 Fig13
Q9	1 / 2	Performance interference measured; security/fault boundaries not tested in checked portion. sections5.2-5.3
Q10	2 / 2	Code, prediction limitations and PCIe caveat explicit. sections4.2,5.2

B.3 S003 / P003 gpulets

[14]

Field	Extracted content
Mechanism	MPS + temporal
Hardware and environment	RTX2080Ti 11GB; two/server; Ubuntu18.04; CUDA10.2; driver440.64; PyTorch1.10
Runtime	`NVIDIA440.64;CUDA10.2;PyTorch1.10` (p207 Table2)
Workload	Nine models; Poisson arrivals; per-model SLO
Tenants	paired local inferences; system scenarios contain several models
Load and SLO	Poisson inter-arrivals; model SLO5-202ms;99% requests within SLO; median of3 runs (pp207-208)
Comparator	Adapted SBP/Nexus; best-fit with/without interference
Finding	Local gpulets support interference-aware provisioning; cluster throughput is a different outcome.
Performance isolation	Interference-aware provisioning with per-model SLO constraints. Fig. 7 reports 90% of measured model pairs below 17.66% interference overhead; this is an across-pair CDF, not request p90/p99.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	20-second reorganization interval; overlap new proxy preparation with active serving (p207 Fig11)
Limitations	Server-level gains cannot be treated as one-GPU utilization.
Source	`https://www.usenix.org/system/files/atc22-choi-seungbeom.pdf`
Evidence location	PDF p10 Table2/3; section5; local profiling section3

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Clear SLO-aware co-location objective. p199 abstract;section1

QA	Score / maximum	Rationale and anchor
Q2	2 / 2	MPS partitions and temporal batching described. pp201-207 sections2-4
Q3	2 / 2	Poisson requests;median of3 repeats;throughput/SLO procedure. pp207-208 section5.1-5.2
Q4	2 / 2	Full `GPU/CPU/OS/driver/runtime` and model tables. p207 Tables2-3
Q5	2 / 2	Ported SBP comparator and Nexus incompatibility justified. p207 section5.1
Q6	2 / 2	Maximum throughput with99%SLO definition;system scale explicit. p208 section5.2
Q7	2 / 2	SLO99% procedure;local overhead CDF population explicit. p203 Fig7;p208 section5.2
Q8	2 / 2	Paired model interference and scheduler interference comparison. p203 Fig7;p208 Fig14
Q9	1 / 2	Performance interference checked;security/fault boundary not established in the checked source evidence. p203 Fig7;section5.2
Q10	1 / 2	Prototype and appendix setup available;replication not attempted. p207 Table2;appendixA

B.4 S004 / P004 PipeSwitch

[8]

Field	Extracted content
Mechanism	Temporal context switching
Hardware and environment	V10016GB or T416GB EC2; PyTorch1.3; CUDA10.1
Runtime	`CUDA10.1;PyTorch1.3.0;torchvision0.4.2;scipy1.3.2;`
Workload	Training batch32; inference8; RN152/InceptionV3/BERT; 100 runs
Tenants	one training task preempted for one inference task; separable inference latency
Load and SLO	`training32;inference8;100-run` averages;Fig6 reports minimum/maximum (p508)
Comparator	`Unoptimized/grouped/per-layer` transfer
Finding	Fig7: inference preemption latency reduced by up to 38.2ms.
Performance isolation	Priority inference preempts training. Table 3 reports 6.01 ms additional mean latency for ResNet152 on V100 under the recorded workload; not a tail-latency or adversarial isolation bound.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	Table3 total extra latency;Table4 first-layer startup overhead;kept separate
Limitations	Average switch/inference latency is not p99; multi-VM throughput separate.
Source	`https://www.usenix.org/system/files/osdi20-bai.pdf`
Evidence location	PDF pp11-13; sections6-6.2; Fig7

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Fine-grained task-switching problem and design explicit. p499 abstract;section1
Q2	2 / 2	Controller,memory daemon,workers and pipeline described. pp504-508 sections4-5
Q3	2 / 2	100-run mean;min/max forFig6;setup described. p508 section6
Q4	1 / 2	Hardware/software `specified;OS/kernel/driver` not established in the checked source evidence. p508 section6
Q5	2 / 2	Same modified PyTorch;ready-model and swapping conditions explicit. p508 section6.1
Q6	2 / 2	Inference throughput and utilization denominator defined. p509 section6.1 Fig6
Q7	2 / 2	End-to-end overhead defined;mean and extrema identified. pp508-509 Table3
Q8	2 / 2	Training/inference co-location;inference metric separated. pp508-509 section6.1
Q9	1 / 2	Process/memory control described;security/fault testing not established in the checked source evidence. pp507-508 implementation
Q10	1 / 2	Versioned stack described;limitations documented;replication not attempted. p508 section6;section7

B.5 S005 / P005 REEF conference

[6]

Field	Extracted content
Mechanism	Reset-based temporal preemption + dynamic kernel padding; NVIDIA binary instrumentation
Hardware and environment	Conference: AMD MI50, 16GB, 60 CUs. Journal companion: MI50 16GB and separate NVIDIA GV100 32GB/80 SMs. Version-specific platforms are not a common configuration.
Runtime	`ROCm4.3.0;ApacheTVM0.8.0;NVIDIA` extension CUDA10.2
Workload	DISB A-E,uniform RT except Poisson E; closed-loop BE; `ResNet152,VGG19,YOLOv3,BERT-base,GPT2` (128 tokens,GPT2 prefill). RT rates50%,100% or10% solo-throughput according to workload; Apollo/SVL trace. Separate LLM case batch1,100 decode tokens,MLC-LLM.
Tenants	DISB `A/B:1RT+1BE;C:1RT+5BE;D/E:5RT+5BE`
Load and SLO	Table2 `RT100/220/100/20/20req/s;D` uniform,E Poisson;BE closed-loop;
Comparator	RT-only; sequential priority; GPU streams. NVIDIA closed-source strata compare priority-enabled Triton/TensorRT and Orion/PyTorch without dynamic padding.
Finding	GV100 PyTorch DISB-C: Orion p99 normalized latency1.17x vs Reef below1.05x; normalized total throughput0.97x vs1.14x(Fig29). TensorRT comparison trades throughput: mean1.21x Triton vs1.11x Reef(Fig28). All ratios retain model-solo normalization; do not mix engines/AMD strata.
Performance isolation	Per-request normalized RT distributions/p99 and separately normalized total throughput; engine/stratum retained
Security boundary	Not established by scheduler priority
Fault boundary	Kernel killing/restart is a scheduling mechanism, not adversarial fault containment
Overhead	Conference REAL workload records RT end-to-end overhead below 2%; individual evidence rows retain workload-specific values. Journal companion comparisons use separate engines and normalization.

Field	Extracted content
Limitations	Conference outcomes and original QA remain OSDI 2022 anchored; journal observations are explicitly labelled companion results. Deterministic/stateless kernel assumption; closed-source comparison omits dynamic padding. Sufficient memory and bandwidth assumptions constrain transfer. Both versions map to one study identity.
Source	`https://www.usenix.org/system/files/osdi22-han.pdf`
Evidence location	Conference: printed pp. 548-549, Sections 7.1-7.2; journal extension: article pp. 1:23-1:24 and 1:31-1:34, Figures 28-29

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Concurrent real-time/best-effort inference objective explicit. p539 abstract;section1
Q2	2 / 2	Reset preemption,queues and padding implementation described. sections3-6;pp547-548
Q3	1 / 2	Benchmark arrivals and baseline methods detailed;repeat/variance protocol not established in the checked source evidence. p548 section7.1
Q4	2 / 2	CPU,GPU,host RAM,OS,runtime,models andclients specified. p548 section7.1 Table2
Q5	2 / 2	RT-only and GPUStreams/SEQ conditions;extensions disclosed. p548 section7.1;p549 section7.3
Q6	2 / 2	Total RT+BE throughput and denominator explicit. p549 section7.2 Fig11
Q7	1 / 2	End-to-end latency and load specified;request percentile not established in the checked source evidence. pp548-549 Table2/Fig11
Q8	2 / 2	MultipleRT/BE clients and multiple mixes directly tested. p548 Table2;p549 section7.2
Q9	1 / 2	Performance isolation tested;no extracted memory/security/fault boundary test. pp548-549 section7.2
Q10	2 / 2	Artifact and idempotence/platform limitations explicit. p540 `repository;section8;appendixA`

B.6 S006 / P006 Paella: Low-latency Model Serving with Software-defined GPU Scheduling

[9]

Field	Extracted content
Mechanism	Temporal software scheduling + MPS comparison
Hardware and environment	TeslaT4; XeonSilver4114/64GB; Linux4.15.0; CUDA11.7; modified TVM0.10
Runtime	`Linux4.15.0;CUDA11.7;TVM0.10.0`
Workload	Eight models; lognormal sigma1.5/2; stabilized measurements
Tenants	Mixed inference; lognormal arrivals sigma1.5/2
Load and SLO	Bursty lognormal arrivals with sigma 1.5/2, eight models; evaluates p99 latency and throughput. A common fixed numerical SLO is not established in the current extract.
Comparator	CUDA single/multiple streams; MPS; Triton23.03; ablations
Finding	p99 latency and throughput under bursty model mixes; Fig12 includes MPS.
Performance isolation	Evaluates p99 latency and throughput under mixed inference loads; Fig. 12 includes MPS. This does not establish a universal latency guarantee.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	Compiler/runtime integration: modified TVM and a custom Triton TVM wrapper. A separate numerical integration/dispatch overhead is not assessed in the retained extract.
Limitations	Custom Triton TVM wrapper; MPS omitted from Fig11; dynamic batching unsupported.
Source	`https://vincen.tl/files/ng23paella.pdf`
Evidence location	PDF pp9-12; section7 Table3/Figs11-12; section8

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Explicit goal. p595 section1
Q2	2 / 2	Mechanism classifiable. pp599-603 sections4-6
Q3	1 / 2	Repetition count unclear. p604 methodology
Q4	2 / 2	Versioned setup. p604 methodology
Q5	1 / 2	Wrapper caveat. p604 Table3/baselines
Q6	2 / 2	Defined throughput. pp604-605 Figs11-12
Q7	2 / 2	p99 specified. pp604-605 Figs11-12
Q8	2 / 2	Mixed jobs tested. p605 Figs12-13
Q9	NA / 0	No security/fault or resource-isolation guarantee appraised; fairness is assessed under Q8; N/A with stated rationale. pp606-607 `discussion;scope-specific` rationale
Q10	2 / 2	Artifact and limits. pp606-607 section8

B.7 S007 / P007 Orion: Interference-aware, Fine-grained GPU Sharing for ML Applications

[10]

Field	Extracted content
Mechanism	CUDA API interception + temporal
Hardware and environment	V10016GB/CUDA10.2; A10040GB/CUDA11.3; PyTorch1.12; Python3.9
Runtime	`PyTorch1.12;Python3.9;CUDA10.2/11.3`
Workload	Inference-inference and inference-training; uniform/Poisson; three repeats
Tenants	High-priority plus best-effort inference;Poisson
Load and SLO	High-priority plus best-effort inference; uniform/Poisson arrivals and three repeats. Model-specific target values are not assessed in the retained extract.
Comparator	MPS; streams; temporal; REEF-N reimplementation
Finding	Interference-aware scheduling compares inference latency and BE throughput.
Performance isolation	Interference-aware scheduling evaluates high-priority inference latency alongside best-effort throughput against MPS, streams, temporal scheduling and REEF-N.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	CUDA interception and scheduling integration; clients in the prototype share a process. A standalone numerical scheduling overhead is not assessed in the retained extract.
Limitations	Prototype clients share a process; REEF-N differs from AMD REEF.
Source	`https://fotstrt.github.io/files/2024-orion.pdf`
Evidence location	PDF pp8-9; sections5.3-6.1; Tables1/3

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Explicit goal. p1075 section1
Q2	2 / 2	Interception specified. pp1079-1082 sections4-5
Q3	2 / 2	Three repetitions. p1082 section6.1
Q4	2 / 2	Versioned setup. p1082 section6.1
Q5	1 / 2	Reimplemented comparator. p1083 baselines
Q6	2 / 2	Defined aggregate metric. p1086 section6.2.3
Q7	2 / 2	p99 and arrivals. p1086 section6.2.3
Q8	2 / 2	Collocations tested. p1086 section6.3
Q9	2 / 2	Trust boundary explicit. p1087 section7 Security
Q10	2 / 2	Artifact and limitations. p1087 section7;artifact appendix

B.8 S008 / P008 GSLICE: Controlled Spatial Sharing of GPUs for a Scalable Inference Platform

[15]

Field	Extracted content
Mechanism	MPS adaptive provisioning
Hardware and environment	One V10016GB; XeonGold6148; kernel4.4.0-150; DPDK19.02; CNTK/PyTorch/TensorRT
Runtime	`Linux4.4.0-150;DPDK19.02;CNTK/PyTorch/TensorRT`
Workload	Two-four inference functions; VGG case SLO50ms; traffic replay
Tenants	Concurrent inference functions;25ms target
Load and SLO	Two to four concurrent inference functions and traffic replay; recorded targets include a 25 ms case and a VGG case with 50 ms SLO. Keep workload-specific targets separate.
Comparator	Standalone functions; MPS provisioning variants
Finding	GUE measures efficiency; shadow instance loading takes about12s in Fig15.
Performance isolation	Adaptive MPS provisioning and measured function latency/efficiency; median-latency results are not a p99 isolation guarantee.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	Shadow-instance loading takes approximately 12 s in Fig. 15; loading time is not steady-state request latency.
Limitations	GUE is not GPU busy percentage; median latency not p99; academic mirror.
Source	`https://www.cse.iitb.ac.in/~{}puru/courses/spring2024-25/downloads/gslice.pdf`
Evidence location	PDF pp8-12; section4; Table3; Fig15

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Inference sharing objective explicit.. PDF p1; printed p492; abstract/introduction
Q2	2 / 2	MPS allocation and adaptive batching described.. PDF pp4-8; section3
Q3	1 / 2	Confidence intervals shown; repeated-run count not specified for retained results.. PDF p9; printed p500; section4.1
Q4	2 / 2	V100 testbed, kernel, frameworks and concurrent models specified.. PDF pp8-9; section4 setup
Q5	2 / 2	TensorRT comparison matches GPU-ready batch execution and states excluded transport costs.. PDF p10; printed p501; section4.3
Q6	2 / 2	GUE defined as an efficacy metric; table values are not busy percentages.. PDF p3; printed p494; PDF p9 Table3
Q7	1 / 2	Median latency and SLO shown; request-tail quantile not established.. PDF p10; section4.4; Figures12a-12b
Q8	2 / 2	Two-to-four-function combinations evaluated.. PDF p9; section4.2; Figure10
Q9	2 / 2	Performance boundary evaluated under allocated GPU shares; no adversarial guarantee inferred.. PDF pp5-7,10-11; sections3.2,4.5
Q10	1 / 2	Implementation and sensitivity described; full reproduction package not established in accessed copy.. PDF pp8-13; sections4-6

B.9 S009 / P009 Serving DNNs like Clockwork: Performance Predictability from the Bottom Up

[11]

Field	Extracted content
Mechanism	Temporal exclusive scheduling
Hardware and environment	V10032GB; DellR740; worker32cores/768GB; two GPUs/host
Runtime	TVM v0.7 compiler and custom managed runtime reported (PDF pp4,7-8); exact OS/driver versions not established in setup.
Workload	Local experiment: one GPU; 15 RN50 copies; 16 clients/model; SLO10-500ms
Tenants	15 ResNet50 instances;16 closed-loop clients/model
Load and SLO	One-GPU local experiment: 15 ResNet50 instances, 16 closed-loop clients per model, SLO range 10-500 ms; Fig. 5 reports SLO-compliant goodput.
Comparator	Clockwork variants; INFaaS on different hardware
Finding	Fig5 measures goodput satisfying SLO, rather than raw request throughput.
Performance isolation	Exclusive temporal scheduling and admission control aim at predictable completion. Goodput counts SLO-compliant requests; no universal cross-workload p99 guarantee is inferred.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	Not assessed as a separate numerical overhead in the retained source extract; controller/worker costs and baseline hardware must not be conflated.
Limitations	Cross-hardware INFaaS comparison is indirect; controller/worker distinction.
Source	`https://www.usenix.org/system/files/osdi20-gujarati.pdf`
Evidence location	PDF p10; section6.1; Fig5

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Predictable multi-model inference objective and target behavior explicit.. PDF p2; printed p443; section1
Q2	2 / 2	Controller scheduling, worker actions and managed model runtime described.. PDF pp6-10; sections4-5
Q3	1 / 2	Section6.6 reports three scaling repetitions; repeated-run count for retained Figure5 is not specified.. PDF pp10-11,14; sections6.1,6.6
Q4	2 / 2	V100 hardware, TVM v0.7 compiler, model set and client configuration are reported.. PDF pp4,7,10; Table1; sections2,6
Q5	1 / 2	INFaaS uses different EC2 machines; authors restrict that contrast to qualitative interpretation.. PDF p10; printed p451; section6.1
Q6	2 / 2	Goodput counts successful requests within their deadline and excludes late requests.. PDF p11; printed p452; section6.1
Q7	2 / 2	SLO range, closed-loop load, latency CDF and batch scheduling explanation are explicit.. PDF pp10-11; Figure5; section6.1
Q8	2 / 2	Concurrent model demand and a protected minor workload tested.. PDF pp10-12; sections6.1-6.3; Figures5-7
Q9	2 / 2	Performance-isolation boundary is measured; dedicated-worker assumptions and security exclusions are explicit.. PDF pp11-12,14-15; sections6.2,7
Q10	2 / 2	Runtime, scheduling, model details and limitations support informed implementation appraisal; no local reproduction claimed.. PDF pp7-10,14-15; sections5,7

B.10 S010 / P010 Colocating ML Inference and Training with Fast GPU Memory Handover (SIRIUS)

[18]

Field	Extracted content
Mechanism	Temporal + MPS memory handover
Hardware and environment	V10016GB; Ubuntu20.04; CUDA11.6; PyTorch2.1.2; TVM0.14; LLM case A10080GB
Runtime	Python3.10; CUDA11.6; PyTorch2.1.2; TVM0.14;
Workload	One GPU; six DNNs/up to56 instances + SWIN training; 300s; Poisson MAF; SLO4x solo
Tenants	six inference models duplicated round-robin to56 instances; co-located Swin-T training, batch72
Load and SLO	Light/Heavy/Burst/Skewed and compressed Azure Functions traces; 300s runs; SLO=4x standalone inference execution; Tidle5s, watermark1GB
Comparator	Reimplemented TaskSwitch; SP50/SP75; UM+MPS
Finding	Inference p99/SLO outcomes accompany training progress; separate TTFT/TBT LLM case.
Performance isolation	Inference p99/SLO compliance evaluated alongside best-effort training progress; SLO is four times standalone inference execution in the recorded trace tests.
Security boundary	Safe memory reclamation/handover is a design property. Adversarial cross-tenant memory isolation is not established by the retained performance evidence.
Fault boundary	Safe memory handover is assessed as runtime correctness; cross-tenant fault containment is not established by the retained evidence.
Overhead	memory adjustment/allocation and batch discard evaluated in section6.3;
Limitations	TaskSwitch is a reimplementation, not vanilla PipeSwitch.
Source	`https://www.usenix.org/system/files/atc25-wang-jiali.pdf`
Evidence location	Section6.1 Fig10; section6.6

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Inference-priority co-location objective and training progress goal explicit.. PDF p2; printed p1657; abstract/introduction
Q2	2 / 2	VMM pool, handover, memory mapping and SM allocation described.. PDF pp5-10; sections3-5
Q3	1 / 2	Trace distributions and 300-second window specified; repeated-run uncertainty not specified for Figure10.. PDF pp10-11; section6; Figure10
Q4	2 / 2	V100/A100 cases, CPU/RAM, software versions and model/batch table supplied.. PDF pp10-11,14; Table3; sections6,6.6
Q5	1 / 2	TaskSwitch is an adapted implementation; equivalence to unmodified PipeSwitch is not established.. PDF p10; comparing targets
Q6	2 / 2	Training samples/second and inference SLO compliance are defined separately.. PDF p11; metrics; Figure10
Q7	2 / 2	P99, arrivals, duration and four-times-solo SLO definition stated.. PDF p11; section6.1
Q8	2 / 2	Inference plus Swin-T training co-location evaluated on a single GPU.. PDF p11; Figure10
Q9	1 / 2	Handover correctness and priority boundary described; direct fault-injection validation not established.. PDF pp6-10; sections4-5
Q10	2 / 2	Artifact appendix reports availability, functionality and reproduction badges with artifact DOI.. PDF p20; artifact appendix; `DOI10.5281/zenodo.15581800`

B.11 S011 / P011 ParvaGPU: Efficient Spatial GPU Sharing for Large-Scale DNN Inference in Cloud Environments

[1]

Field	Extracted content
Mechanism	MIG + MPS
Hardware and environment	A10080GB; Ubuntu20.04; CUDA12.0.1; reported PyTorch1.14.0 needs checking
Runtime	CUDA12.0.1; PyTorch1.14.0 as printed in source (unverified version string);
Workload	Eleven models; six scenario mixes; same-model MPS inside MIG
Tenants	one to three homogeneous MPS processes within a MIG instance; other models can occupy separate segments
Load and SLO	extracted example:1-GPC MIG instance; batch4;1/2/3 MPS processes; no tail quantile identified for local latency
Comparator	MIG-serving; gpulet and system variants
Finding	Fig7 separates internal/external fragmentation; Fig5 counts GPUs.
Performance isolation	MIG/MPS configurations evaluated against workload-specific SLOs and GPU usage; local batch-4 latency example does not identify a tail percentile.
Security boundary	MIG partitioning is an architectural boundary; a dedicated adversarial security assessment is not established in the retained extract.
Fault boundary	Cross-tenant fault containment is not assessed in the retained source extract; GPU counts and SLO results do not establish it.
Overhead	scheduling/profiling discussed;
Limitations	Author manuscript; unusual runtime version; GPU count is not occupancy.
Source	`https://arxiv.org/pdf/2409.14447`
Evidence location	PDF p8 onward; sectionIV; TableIV; Figs5/7

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	SLO-constrained inference resource objective explicit.. arXiv v1 PDF pp1-2
Q2	2 / 2	MIG/MPS profiling, segment configuration and allocation algorithms described.. arXiv v1 PDF pp4-8; sectionIII
Q3	1 / 2	Profiler and six scenarios specified; repeated-run uncertainty not established.. arXiv v1 PDF pp5,8-10; sectionsIII-C,IV
Q4	1 / 2	A100, Ubuntu and CUDA reported; printed PyTorch1.14.0 string is an unresolved environment ambiguity.. arXiv v1 PDF p8; sectionIV-A
Q5	1 / 2	Local process-count contrast interpretable; fleet baseline variants and manuscript scope constrain comparison.. arXiv v1 PDF pp5,8; sectionsIII-B,IV-A
Q6	2 / 2	Local throughput values and metric units given.. arXiv v1 PDF pp4-5; Figure3; sectionIII-B
Q7	1 / 2	Local latency values given without a named tail quantile.. arXiv v1 PDF pp4-5; Figure4; sectionIII-B
Q8	2 / 2	One-to-three MPS processes measured inside a MIG segment.. arXiv v1 PDF pp4-5; Figures3-4
Q9	1 / 2	MIG/MPS boundary described; no dedicated fault or adversarial experiment established in reviewed manuscript.. arXiv v1 PDF pp2-4,8-10
Q10	1 / 2	Algorithms and reconfiguration limitation described; accessed manuscript has no verified final-version equivalence.. arXiv v1 PDF pp5-10; sectionIII-F; source/version record

B.12 S012 / P012 Torpor: GPU-Enabled Serverless Computing for Low-Latency, Resource-Efficient Inference

[19]

Field	Extracted content
Mechanism	API remoting + temporal swapping
Hardware and environment	Worker48vCPU/384GB; four V10032GB
Runtime	PyTorch-based clients replace CUDA libraries; exact OS/driver/CUDA versions not reported in section7 settings
Workload	Eight inference models; warm-up; CV80ms/BERT200ms SLO
Tenants	multiple low-frequency ResNet152 functions share one GPU; sufficient functions to saturate device memory
Load and SLO	10-80 `requests/minute/function;` global defaultP98; CV deadline80ms and Bert-qa200ms
Comparator	Native; remote GPU; PCIe/NVLink swap
Finding	BERT Table4: native42ms, remote43ms, PCIe swap144ms, NVLink swap45ms.
Performance isolation	P98 response-time/SLO results evaluated under low-frequency serverless inference, with CV 80 ms and BERT-QA 200 ms deadlines; remoting and swapping conditions remain separate.
Security boundary	Section4.5 describes shared and isolated GPU runtimes; section7 evaluates runtime sharing, while section8 deployment uses isolation mode. No universal adversarial-security guarantee inferred.
Fault boundary	Section4.5 describes restart, migration and persisted-state recovery. Direct fault-injection validation is not established by the retained performance results.
Overhead	API redirection, swapping and extra CPU activity evaluated; ResNet152 remoting17ms versus Native25ms, Table4
Limitations	p98 tail; busy time is not SM occupancy; node scaling separate.
Source	`https://www.usenix.org/system/files/atc25-yu.pdf`
Evidence location	PDF pp10-11; Table4; sections7.2-7.3

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Low-frequency serverless inference density and latency objectives explicit.. PDF p2; printed p597; section1
Q2	2 / 2	Remoting, swapping, memory manager, scheduler and two runtime modes described.. PDF pp5-9; sections4-6
Q3	1 / 2	Trace, rates and warm-up specified; repeated-run uncertainty not specified for retained Figure8.. PDF pp9-11; section7
Q4	1 / 2	Worker and model configuration supplied; exact OS/driver/framework version matrix not specified in setup.. PDF p9; printed p604; settings
Q5	1 / 2	Native and adapted INFless-KA baselines described; CPU execution paths and adaptations affect equality.. PDF pp10-11; sections7.1-7.3
Q6	2 / 2	Normalized single-GPU throughput and GPU busy-time definition supplied.. PDF pp10-11; metrics; Figure8
Q7	2 / 2	Default P98, per-function rates and deadlines stated.. PDF pp10-11; metrics; section7.2
Q8	2 / 2	Many low-frequency functions share one GPU; node-level results identified separately.. PDF pp10-11; Figure8; section7.2
Q9	1 / 2	Runtime-sharing and runtime-isolation modes plus recovery design stated; dedicated fault-injection evaluation not established.. PDF p7; section4.5; PDF p9 evaluation mode
Q10	1 / 2	Implementation and mode trade-offs explained; complete public replication recipe not established in paper.. PDF pp7-9,13; sections4.5,6,8

B.13 S013 / P013 StreamBox: A Lightweight GPU SandBox for Serverless Inference Workflow

[20]

Field	Extracted content
Mechanism	API/runtime sandbox virtualization
Hardware and environment	Four V10016GB; 28cores/128GB; CUDA10.1; PyTorch1.3; TensorFlow2.12
Runtime	`PyTorch1.3.0,TensorFlow2.12,CUDA10.1` as reported; driver/OS version not reported in section9 setup
Workload	Three concurrent inference workflows/GPU
Tenants	concurrent functions within workflow share runtime; separate3/6-user Traffic test on one GPU
Load and SLO	Azure Functions7-day invocation trace; `sporadic/periodic/bursty;` Traffic SLO200ms; up to3 workflows/runtime contexts per GPU in evaluated setup
Comparator	MPS; INFless; Astraea; Stream-only
Finding	Table2 evaluates runtime sandbox workflows on shared GPUs.
Performance isolation	Concurrent workflow/runtime configurations evaluated with traffic SLO 200 ms and separate 3/6-user tests. This is workload-specific performance evidence.
Security boundary	Runtime sandbox memory separation is discussed; the retained extract does not establish adversarial security.
Fault boundary	Fault-containment behavior is not assessed in the retained source extract; sandbox terminology alone is not a fault test.
Overhead	API `forwarding1-3microseconds;3-5microseconds` at48functions; one-time profiling36s and transformation4s, p70
Limitations	Sandbox memory separation is not proof of adversarial security.
Source	`https://www.usenix.org/system/files/atc24-wu-hao.pdf`
Evidence location	PDF p10 onward; section9; Table2

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Workflow runtime-redundancy and spatial-sharing objective explicit.. PDF p2; printed p59; section1
Q2	2 / 2	API proxy, PTB transformation, stream sharing and memory pool described.. PDF pp5-10; sections3-8
Q3	1 / 2	Trace types, ablations and workloads stated; repeated-run uncertainty not specified for Figure13.. PDF pp10-13; section9
Q4	2 / 2	V100 hardware, CPU/RAM, framework versions, CUDA version and workflow descriptions reported.. PDF p10; printed p67; setup; Table2
Q5	1 / 2	Warm-up and common optimizations disclosed; adapted baselines and batch/concurrency choices constrain causal attribution.. PDF pp10-11; comparing targets; Figure13
Q6	2 / 2	Throughput, memory metrics and comparator-specific factors explicitly reported.. PDF p11; printed p68; Figures12-13
Q7	1 / 2	Traffic SLO and CDF supplied; exact scalar request-tail quantile not supplied for retained throughput result.. PDF p11; Figure14
Q8	2 / 2	Separate three- and six-user single-GPU tests supplied.. PDF p13; section9.4; Figure19
Q9	2 / 2	Mutual trust within workflow is explicit; separate proxies for users are distinguished from untested strong security.. PDF pp5-6,13; sections3-4,9.4,10
Q10	2 / 2	Implementation, preparation costs and multi-user limitations sufficiently detailed for informed appraisal.. PDF pp6-10,13; sections4-9

B.14 S014 / P014 Efficient Performance-Aware GPU Sharing with Compatibility and Isolation through Kernel Space Interception (KRYPTON)

[21]

Field	Extracted content
Mechanism	Kernel API virtualization + hybrid
Hardware and environment	A10040GB/RTX4090; Ubuntu20.04; kernel5.15.0-91; driver550.90.07; CUDA12.1; PyTorch2.2.1
Runtime	`NVIDIA550.90.07;Docker24.0.7;CUDA12.1;Vulkan1.3;PyTorch2.2.1`
Workload	`Inference/training/rendering` in separate containers; solo-throughput targets
Tenants	inference with co-running applications; separate training/rendering results not substituted
Load and SLO	QoS: initial Poisson rate is half solo maximum; shared setting halves the input again and assigns50%time; tokens5/10/20/40/80ms; target original soloP99.
Comparator	Temporal; MIG; adapted gpulet
Finding	Inference-specific sharing comparisons; section2.2 tests fault propagation.
Performance isolation	Inference QoS uses solo-p99 targets under Poisson load at half maximum throughput; temporal allocation/token settings and adapted baselines are retained.
Security boundary	Container/kernel interception and partition boundaries are design features. A dedicated adversarial security assessment is not established by the retained extract.
Fault boundary	Section 2.2 Fig. 3 examines fault propagation in sharing baselines. This observation is not treated as proof of universal KRYPTON fault containment.
Overhead	token-length trade-off and4.8MBhost-memory module overhead
Limitations	Baseline adaptations and workload mix constrain attribution.
Source	`https://www.usenix.org/system/files/atc25-zhang-shulai.pdf`
Evidence location	PDF p10 Tables1-2; section5; section2.2 Fig3

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Compatibility, isolation and performance-control objectives explicit.. PDF p2; printed p1003; section1
Q2	2 / 2	Kernel interception, token control and MIG orchestration explained.. PDF pp5-10; sections3-4
Q3	1 / 2	Setup and workloads detailed; repeated-run uncertainty not specified for retained QoS result.. PDF pp10-13; section5
Q4	2 / 2	OS, kernel, driver, runtimes, GPUs and workloads explicitly tabulated.. PDF p10; printed p1011; Tables1-2
Q5	1 / 2	Isolation-capable baselines and MIG adaptations described; policy and allocation differences limit equivalence.. PDF pp10-11; section5.1
Q6	2 / 2	Throughput targets are solo-normalized and QoS violation is a request fraction.. PDF pp10,12-13; sections5.1,5.4-5.6
Q7	2 / 2	Solo-P99 threshold, Poisson load, subsequent rate halving and token sizes stated.. PDF pp12-13; section5.5; Figure17
Q8	2 / 2	Co-running inference relative-error and QoS experiments supplied.. PDF p13; sections5.5-5.6
Q9	1 / 2	Baseline fault propagation is tested; design/security argument does not directly establish universal KRYPTON containment.. PDF p4 Figure3; PDF p14 section7
Q10	1 / 2	Implementation and MIG limitation detailed; complete end-to-end replication package not established.. PDF pp10,14; sections4,7

B.15 S015 / P015 XSched: Preemptive Scheduling for Diverse XPUs

[2]

Field	Extracted content
Mechanism	Temporal API interposition
Hardware and environment	One GV100 inference case; broader XPU experiments separate
Runtime	CUDA/Triton and Paella artifact; exact version numbers not reported in extracted case3
Workload	Two BERT-large clients; HP10req/s; saturated BE
Tenants	one high-priority client `at10requests/s;low-priority` client continuous on sameGV100
Load and SLO	Triton client rates as above;Paella workload lognormal sigma2.0;K-EDF K16; exact batch sizes not reported in extracted case3
Comparator	Triton; Triton+priority; Triton+XSched
Finding	Fig15: relative p99 1.53/1.51/1.07 respectively versus solo.
Performance isolation	Two BERT-large clients on one GV100: Fig. 15 gives relative p99 1.53/1.51/1.07 for `Triton/Triton+priority/Triton+XSched` versus solo.
Security boundary	Scheduling API interposition does not establish adversarial memory isolation; security is not assessed in this extracted inference case.
Fault boundary	Cross-tenant fault containment is not assessed in the extracted inference case; device/preemption limitations remain separate.
Overhead	10lines Triton integration and15lines Paella-baseline integration;preemption overhead measured separately
Limitations	Data assumed to fit memory; CUDA graphs and undocumented driver limits.
Source	`https://www.usenix.org/system/files/osdi25-shen-weihang.pdf`
Evidence location	PDF pp14-15; section8 Case3 Fig15; section9

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Portable scheduling objective andGPUcase clear.. pp671-672 section1
Q2	2 / 2	XQueue abstraction andmulti-level preemption described.. pp674-679 sections3-6
Q3	1 / 2	Workload rates and policies explicit; repeat uncertainty not established.. pp680-683 section7
Q4	1 / 2	GPU/model/rates explicit;host/software versions incomplete in extracted case.. p683 case3
Q5	2 / 2	Same-GPU standalone/native priority comparisons supplied.. p683 Figure15
Q6	2 / 2	Throughput-latency curves defined forGPUinferencecase.. p683 Figure15b
Q7	2 / 2	P99 and offered rates explicit.. p683 Figure15a
Q8	2 / 2	Two concurrentBertclients ononeGV100.. p683 case3
Q9	NA / 0	No claimed `security/fault-containment` guarantee in extracted scheduling claim;N/A with stated rationale for claim-specificQ9.. pp671-672 scope; p683 case3
Q10	2 / 2	`Implementation/interfaces,artifact` andhardware-dependence detailed.. sections6-8;artifact appendix

B.16 S016 / P018 InferFair: Towards QoS-aware scheduling for performance isolation guarantee in heterogeneous model serving systems

[24]

Field	Extracted content
Mechanism	Temporal SLO-aware scheduling + feedback goodput balancing; MPS comparator
Hardware and environment	Dell PowerEdgeR740; V10032GB; local synthetic co-running experiment and separate four-GPU realistic trace.
Runtime	Not assessed in the retained source extract; exact software versions remain unverified.
Workload	Local GoogLeNet/ResNet152 Poisson arrivals; three-model test adds `Mobile_Pose_MobileNetv3;600req/s` distributed in six ratios; SLO100ms and interval50ms. Separate61-model,90min MAF replay on four GPUs.
Tenants	Local two-model GoogLeNet/ResNet152 test; three-model test adds `Mobile_Pose_MobileNetv3.` Four-GPU 61-model replay is a separate experiment.
Load and SLO	Poisson arrivals; three-model aggregate 600 requests/s split across six ratios; SLO 100 ms; feedback interval 50 ms. Separate MAF replay lasts 90 min.
Comparator	Clockwork; MPS plus Clockwork SLO scheduling with two worker processes/GPU; EDF without batching (not a like-for-like batching baseline).
Finding	Local Table3 load3:1:2: unfairness1.05 vs Clockwork1.50; SLO goodput524.29 vs457.24req/s. Other load ratios give smaller differences. Local unfairness is max/min model slowdown(Eq2), not p99 or security. Four-GPU MAF outcome kept separate.
Performance isolation	Relative goodput/slowdown fairness and SLO-compliant goodput; no extracted p99 bound
Security boundary	Not established by performance-isolation terminology
Fault boundary	Not tested in extracted results
Overhead	Online feedback/scheduling and batching costs are qualitative here; a standalone numerical scheduling overhead is not assessed in the retained extract.

Field	Extracted content
Limitations	Two-column Markdown interleaves algorithms/prose; use labelled Table3 rows and sections, not scrambled formulas. Repetitions/variance and software versions unresolved. MAF trace has minute-level invocation counts; cross-study batching and isolation definitions differ.
Source	`https://www.sciencedirect.com/science/article/pii/S0167739X23003187`
Evidence location	Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Heterogeneous inference fairness is explicit. Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q2	2 / 2	Temporal scheduler and feedback controller are described. Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q3	1 / 2	Configuration/control details are incomplete in accessible extraction. Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q4	2 / 2	Models, arrival rates and SLOs are explicit. Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q5	1 / 2	EDF lacks batching and is not a matched baseline. Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q6	2 / 2	Goodput and unfairness retain separate definitions. Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q7	1 / 2	Service SLOs are given; request-tail quantile evidence is limited. Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q8	2 / 2	Multiple concurrent inference models are evaluated. Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q9	1 / 2	Performance fairness does not establish security/fault separation. Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899
Q10	1 / 2	Algorithms and limits are interpretable; software-version coverage is incomplete. Published pp17-18; sections5.1-5.4; Table3; Eq2; Markdown lines718-899

B.17 S017 / P019 ArkGPU: enabling applications' high-goodput co-location execution on multitasking GPUs

[22]

Field	Extracted content
Mechanism	CUDA API interception + NVML monitoring and admission throttling
Hardware and environment	Machine with four V100s; Xeon Platinum8163; CentOS7.9.2009; driver460.106.00; CUDA11.1; cuDNN7.6.5.
Runtime	CentOS 7.9.2009; NVIDIA driver 460.106.00; CUDA 11.1; cuDNN 7.6.5 (recorded setup).
Workload	Five MLPerf inference LC models (ResNet50, SSD-MobileNet, SSD-ResNet34, DLRM, BERT), each paired locally with six BE training models (30 pairs). Single-stream query size1; p99 QoS threshold twice solo. Separate four-GPU app-mixes are not local occupancy evidence.
Tenants	One latency-critical inference model paired with one best-effort training model locally; five inference by six training models give 30 tested pairs.
Load and SLO	MLPerf single-stream, query size 1; inference QoS threshold is twice solo p99. Separate multi-GPU app mixes are not the local test.
Comparator	MPS (Non-ArkGPU); k8s-native for QPS/utilization; GaiaGPU and adjustment-unit ablation.
Finding	Section5.2 reports all30 tested pairs satisfy the 2x-solo p99 QoS target with ArkGPU. Low-contention pairs can lose about5% QPS from throttling. Four-GPU app-mix goodput gains94.98% vs k8s-native and38.65% vs MPS are separate system-level results.
Performance isolation	p99 tested against twice-solo QoS; average latency, QPS and goodput separately
Security boundary	Not established by API interception
Fault boundary	Background discussion of MPS faults is not an ArkGPU fault test
Overhead	Polling/API sleeps can add delay; BE checkpoint/restart incurs costs. Low-contention pairs can lose about 5% QPS; this is a measured effect, not a universal implementation-overhead figure.

Field	Extracted content
Limitations	Mixed inference/training qualifies via explicit inference p99/QPS and workload transfer rationale; it does not establish inference-only performance. Polling/API sleeps can add delay; BE checkpoint/restart affects costs. Huge goodput ratios can have near-zero baseline yield. Repetitions/variance and chart coordinates unresolved.
Source	`https://doi.org/10.1007/s42514-023-00154-y`
Evidence location	Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Latency-critical inference under co-location is explicit. Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q2	2 / 2	Interception, NVML and throttling are specified. Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q3	1 / 2	Repeated-run precision is not established by the available source assessment. Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q4	2 / 2	GPU, software and thirty model pairs are described. Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q5	1 / 2	Exact comparator MPS quotas are not established. Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q6	2 / 2	QPS and goodput are distinguished. Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q7	2 / 2	p99 criterion is twice solo latency. Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q8	2 / 2	Thirty inference/training pairs are tested. Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360

QA	Score / maximum	Rationale and anchor
Q9	1 / 2	No adversarial or cross-tenant fault test is established. Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360
Q10	1 / 2	Overhead and low-contention regression are discussed; replication gaps remain. Published pp313-319; sections4.3,5.1-5.4; Table3; Figs14-21; Markdown lines920-1360

B.18 S018 / P020 Dyna-P: placement-aware dynamic partitioning for lightweight applications with modern GPUs

[16]

Field	Extracted content
Mechanism	MIG dynamic `placement/reconfiguration` with reconfiguration-last policy
Hardware and environment	Two A30s,24GB each; CUDA12.2; driver535.171.04; DCGM3.1.3. Local sections4.2/4.3.1 use one A30. StaCon `profiles2g.12gb+1g.6gb+1g.6gb.`
Runtime	CUDA 12.2; NVIDIA driver 535.171.04; DCGM 3.1.3 (recorded setup).
Workload	50 SLM inference jobs; batch1-64; DistilGPT, Codegen, Flan-T5-Large, Codellama and Phi-1 as labelled in Table2. Local batch sensitivity:10 inference jobs, batches1-8 vs32-64 on one A30. Later160-A100 cluster study is simulation.
Tenants	Multiple SLM inference jobs occupy MIG partitions; local batch experiment has 10 jobs on one A30. Separate 50-job and simulated cluster results are distinguished.
Load and SLO	Inference batch sizes 1-64; local sensitivity compares batches 1-8 versus 32-64. JCT/makespan/STP are job metrics, not online request-tail SLO evidence.
Comparator	Full-GPU sequential baseline; fixed StaCon MIG; pre-profiled MISO adapted to A30; ORACLE with future job/size/time knowledge.
Finding	Single-A30 batch experiments: smaller partitions increase concurrency but can increase mean JCT; not online p99 evidence. Fig8 broader job set reports39.03% lower makespan vs StaCon,12.14% vs MISO and14.7% higher normalized STP vs StaCon. Prediction-error sensitivity:15% perturbation adds8.09% makespan.
Performance isolation	Job completion/makespan/STP and prediction sensitivity; no extracted online p99 guarantee
Security boundary	Architectural isolation discussed, no adversarial test
Fault boundary	OOM observation is not cross-tenant fault containment validation
Overhead	Reconfiguration may stop/restart active jobs; reconfiguration-last placement reduces interruption. A reliable numerical overhead fraction is not assessed from the retained extract.

Field	Extracted content
Limitations	Retain one-GPU inference strata separately from two-GPU aggregate and cluster simulation. JCT/makespan/STP cannot replace request-tail latency or goodput. Reconfiguration can stop/restart active jobs; neither architectural MIG claims nor OOM avoidance prove hostile isolation. Model labels/parameter counts and corrupted overhead fraction require caution.
Source	`https://doi.org/10.1007/s10586-025-05284-2`
Evidence location	PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Fragmentation and concurrent inference are explicit. PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q2	2 / 2	MIG placement and merge/split policies are described. PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q3	1 / 2	Repetition/variance detail is limited in accessible evidence. PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q4	2 / 2	GPU profiles and local inference job ranges are specified. PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q5	1 / 2	MISO adaptation and privileged ORACLE require caveats. PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q6	2 / 2	Makespan and STP are separate metrics. PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q7	1 / 2	JCT is not online request p99. PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q8	2 / 2	Local concurrent inference job strata are identifiable. PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865
Q9	1 / 2	Architectural MIG descriptions do not constitute adversarial tests. PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865

QA	Score / maximum	Rationale and anchor
Q10	1 / 2	Reconfiguration and prediction limits are explicit; software coverage is incomplete. PDF pp11-16; sections4.1-4.3.2; Table2; Figs8-13; Markdown lines700-865

B.19 S019 / P021 KRISP: Enabling Kernel-wise RIght-sizing for Spatial Partitioned GPU Inference Servers

[17]

Field	Extracted content
Mechanism	Spatial partition/API control (AMD)
Hardware and environment	One MI50/60CUs; dualEPYC7302/512GB; Ubuntu18.04; kernel5.4; ROCm4.5
Runtime	ROCm 4.5; Ubuntu 18.04; kernel 5.4 (recorded setup).
Workload	One/two/four workers and heterogeneous pairs; batch32; aggregate8/16; eight models
Tenants	One, two and four workers; heterogeneous model pairs on one AMD MI50.
Load and SLO	Recorded tests include batch 32 and aggregate batches 8/16; throughput and p95 inference latency are evaluated. A request-rate/SLO target is not established in this extract.
Comparator	AMD unrestricted concurrency labelled MPSDefault
Finding	Kernel-wise right-sizing evaluates throughput and p95 inference latency.
Performance isolation	Kernel-wise CU right-sizing evaluated through throughput and p95 inference latency. AMD unrestricted concurrency is not NVIDIA MPS.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	The feature is emulated and emulation overhead is subtracted in reported comparisons. Do not interpret adjusted throughput as measured deployable runtime cost.
Limitations	Feature emulated; overhead subtracted; baseline is not NVIDIA MPS.
Source	`https://danielwong.org/files/KRISP-HPCA2023.pdf`
Evidence location	PDF pp9-10; sectionsV-VI; TableIII

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Kernel-level right-sizing objective is explicit. PDF pp9-10; sectionsV-VI; TableIII
Q2	2 / 2	API control and spatial sizing are classifiable. PDF pp9-10; sectionsV-VI; TableIII
Q3	1 / 2	Feature emulation limits direct experimental transfer. PDF pp9-10; sectionsV-VI; TableIII
Q4	2 / 2	MI50 and tested inference workers/batches are specified. PDF pp9-10; sectionsV-VI; TableIII
Q5	1 / 2	MPSDefault is AMD unrestricted concurrency, not NVIDIA MPS. PDF pp9-10; sectionsV-VI; TableIII
Q6	2 / 2	Throughput is defined for the tested worker configuration. PDF pp9-10; sectionsV-VI; TableIII
Q7	2 / 2	p95 inference latency is evaluated. PDF pp9-10; sectionsV-VI; TableIII
Q8	2 / 2	Concurrent inference workers and heterogeneous pairs are tested. PDF pp9-10; sectionsV-VI; TableIII
Q9	1 / 2	Resource sizing is not a validated security/fault boundary. PDF pp9-10; sectionsV-VI; TableIII
Q10	1 / 2	Emulation and subtracted overhead limit deployability. PDF pp9-10; sectionsV-VI; TableIII

B.20 S020 / P022 Tally: Non-Intrusive Performance Isolation for Concurrent Deep Learning Workloads

[23]

Field	Extracted content
Mechanism	CUDA API interception + priority
Hardware and environment	A10040GB; Ubuntu20.04; CUDA12.2; PyTorch2.2; ONNX1.17
Runtime	Ubuntu 20.04; CUDA 12.2; PyTorch 2.2; ONNX 1.17 (recorded setup).
Workload	Same-GPU pairs; MAF inference traces; high-priority p99
Tenants	High-priority inference and best-effort training workloads share one A100; model/workload combinations remain experiment-specific.
Load and SLO	MAF-derived inference arrival traces; high-priority p99 compared with standalone execution. The retained throughput contrast uses a 10% idle-time condition.
Comparator	Time slicing; MPS; priority MPS; TGS
Finding	At 10% idle time: normalized throughput Tally1.19 vs TGS1.65.
Performance isolation	Thread-block-level scheduling evaluates high-priority inference p99 alongside best-effort throughput; latency protection is not security or fault isolation.
Security boundary	Not assessed for adversarial security in the retained source evidence; performance-sharing results do not establish memory/security isolation.
Fault boundary	Not assessed for cross-tenant fault containment in the retained source evidence; scheduling/performance results do not establish it.
Overhead	At 10% idle time normalized throughput is 1.19 for Tally versus 1.65 for TGS; this is a throughput trade-off, not a standalone integration-overhead measurement.
Limitations	Latency protection sacrifices some BE throughput; author manuscript.
Source	`https://arxiv.org/pdf/2410.07381`
Evidence location	PDF pp9-11; sections5-5.3; Fig6

QA	Score / maximum	Rationale and anchor
Q1	2 / 2	Performance isolation during co-location is explicit. PDF pp9-11; sections5-5.3; Fig6
Q2	2 / 2	CUDA interception and priority control are described. PDF pp9-11; sections5-5.3; Fig6
Q3	1 / 2	Repeated-run precision is not established in the checked source summary. PDF pp9-11; sections5-5.3; Fig6
Q4	2 / 2	GPU and principal software versions are specified. PDF pp9-11; sections5-5.3; Fig6
Q5	1 / 2	TGS and scheduling policies trade different background yields. PDF pp9-11; sections5-5.3; Fig6
Q6	2 / 2	Foreground protection and background throughput remain distinct. PDF pp9-11; sections5-5.3; Fig6
Q7	2 / 2	High-priority p99 is an evaluated outcome. PDF pp9-11; sections5-5.3; Fig6
Q8	2 / 2	Same-GPU co-tenant pairs and production-style traces are used. PDF pp9-11; sections5-5.3; Fig6
Q9	1 / 2	Priority control does not establish hostile memory/fault isolation. PDF pp9-11; sections5-5.3; Fig6
Q10	2 / 2	Artifact, implementation and throughput sacrifice support interpretation. PDF pp9-11; sections5-5.3; Fig6

C Complete outcome register

Reported observations remain attached to their baseline, hardware, workload and source. Values are preserved as supplied, including ranges, approximate values and maxima.

E001 / P001 Throughput ratio

Field	Evidence
Value	2.7
Units	x
Baseline	No MPS
Hardware	NVIDIA A100
Workload	VGG16+GoogLeNet
Interpretation	Batch/rate/quotas unresolved; not tail-latency evidence
Evidence location	pp457-458 Fig14 and accompanying prose
Source	s11390-024-3773-5.md

E002 / P001 Throughput ratio

Field	Evidence
Value	1.5
Units	x
Baseline	Default MPS
Hardware	NVIDIA A100
Workload	VGG16+GoogLeNet
Interpretation	Same comparison family; not a universal average
Evidence location	pp457-458 Fig14 and accompanying prose
Source	s11390-024-3773-5.md

E003 / P002 Prediction error range

Field	Evidence
Value	1.53-5.02
Units	percent
Baseline	Measured local latency
Hardware	NVIDIA V100
Workload	`AlexNet/ResNet50/VGG19/SSD;` batch3
Interpretation	Prediction accuracy, not utilization or latency speedup
Evidence location	p822 section5.2 Fig13
Source	iGniter user-supplied Markdown

E004 / P002 Initial p99 latency

Field	Evidence
Value	15.6
Units	ms
Baseline	W1 latency SLO10ms
Hardware	NVIDIA V100
Workload	W1 with W5 and W11
Interpretation	Transient violation precedes recovery; no unconditional guarantee
Evidence location	p823 section5.3 Fig17 prose
Source	iGniter user-supplied Markdown

E005 / P004 Additional end-to-end latency

Field	Evidence
Value	6.01
Units	ms
Baseline	Ready-model latency subtracted; MPS overhead307.02ms
Hardware	NVIDIA V10016GB
Workload	ResNet152 inference batch8 + training batch32

Field	Evidence
Interpretation	Mean100 runs; overhead, not total latency or p99; MPS uses unified-memory swapping
Evidence location	p509 Table3; setup p508 section6
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E006 / P003 Distribution threshold:90% of measured pairs

Field	Evidence
Value	17.66
Units	percent overhead below threshold
Baseline	Solo-running latency
Hardware	RTX2080Ti11GB
Workload	Paired model inferences
Interpretation	Across-pair CDF, not request-p90/p99; no hostile-memory/fault test
Evidence location	p203 Fig7
Source	https://www.usenix.org/system/files/atc22-choi-seungbeom.pdf

E007 / P004 Additional end-to-end latency

Field	Evidence
Value	307.02
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA V10016GB
Workload	ResNet152 inference batch8 + training batch32
Interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Evidence location	p509 Table3; setup p508 section6
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

Field	Evidence

E008 / P004 Additional end-to-end latency

Field	Evidence
Value	5.40
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA V10016GB
Workload	Inception_v3 inference batch8 + training batch32
Interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Evidence location	p509 Table3; setup p508 section6
Source	`https://www.usenix.org/system/files/osdi20-bai.pdf`

E009 / P004 Additional end-to-end latency

Field	Evidence
Value	232.25
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA V10016GB
Workload	Inception_v3 inference batch8 + training batch32
Interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Evidence location	p509 Table3; setup p508 section6
Source	`https://www.usenix.org/system/files/osdi20-bai.pdf`

E010 / P004 Additional end-to-end latency

Field	Evidence
Value	10.27
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA V10016GB
Workload	Bert_base inference batch8 + training batch32
Interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Evidence location	p509 Table3; setup p508 section6
Source	`https://www.usenix.org/system/files/osdi20-bai.pdf`

E011 / P004 Additional end-to-end latency

Field	Evidence
Value	204.52
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA V10016GB
Workload	Bert_base inference batch8 + training batch32
Interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Evidence location	p509 Table3; setup p508 section6
Source	`https://www.usenix.org/system/files/osdi20-bai.pdf`

E012 / P004 Additional end-to-end latency

Field	Evidence
Value	5.57
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA T416GB

Field	Evidence
Workload	ResNet152 inference batch8 + training batch32
Interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Evidence location	p509 Table3; setup p508 section6
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E013 / P004 Additional end-to-end latency

Field	Evidence
Value	259.20
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA T416GB
Workload	ResNet152 inference batch8 + training batch32
Interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Evidence location	p509 Table3; setup p508 section6
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E014 / P004 Additional end-to-end latency

Field	Evidence
Value	7.66
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA T416GB
Workload	Inception_v3 inference batch8 + training batch32
Interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Evidence location	p509 Table3; setup p508 section6

Field	Evidence
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E015 / P004 Additional end-to-end latency

Field	Evidence
Value	193.05
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA T416GB
Workload	Inception_v3 inference batch8 + training batch32
Interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Evidence location	p509 Table3; setup p508 section6
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E016 / P004 Additional end-to-end latency

Field	Evidence
Value	34.56
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA T416GB
Workload	Bert_base inference batch8 + training batch32
Interpretation	Mean of100 runs; overhead not total latency or tail statistic; mixed training/inference workload.
Evidence location	p509 Table3; setup p508 section6
Source	https://www.usenix.org/system/files/osdi20-bai.pdf

E017 / P004 Additional end-to-end latency

Field	Evidence
Value	338.25
Units	ms
Baseline	Ready-model latency subtracted
Hardware	NVIDIA T416GB
Workload	Bert_base inference batch8 + training batch32
Interpretation	Training task occupies GPU memory and continues. Baseline condition matters; not a universal MPS comparison.
Evidence location	p509 Table3; setup p508 section6
Source	`https://www.usenix.org/system/files/osdi20-bai.pdf`

E018 / P005 Overall throughput ratio

Field	Evidence
Value	1.60
Units	x
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE
Interpretation	Includes RT and BE requests; RT-only reference dedicates GPU to RT tasks.
Evidence location	p549 section7.2 Fig11 prose; setup p548 Table2
Source	`https://www.usenix.org/system/files/osdi22-han.pdf`

E019 / P005 RT end-to-end latency overhead

Field	Evidence
Value	0.5
Units	percent
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB

Field	Evidence
Workload	DISB A:1 VGG19 RT client100req/s uniform +1 ResNet152 BE
Interpretation	Described as negligible; not a p99 value or worst-case guarantee.
Evidence location	p549 section7.2 Fig11 prose; setup p548 Table2
Source	`https://www.usenix.org/system/files/osdi22-han.pdf`

E020 / P005 Overall throughput ratio

Field	Evidence
Value	1.14
Units	x
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE
Interpretation	High RT load; differs from low-load DISB A.
Evidence location	p549 section7.2 Fig11 prose; setup p548 Table2
Source	`https://www.usenix.org/system/files/osdi22-han.pdf`

E021 / P005 RT end-to-end latency overhead

Field	Evidence
Value	1
Units	percent
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	DISB B:1 VGG19 RT client220req/s uniform +1 ResNet152 BE
Interpretation	About60 microseconds in prose; not a tail percentile.
Evidence location	p549 section7.2 Fig11 prose; setup p548 Table2

Field	Evidence
Source	https://www.usenix.org/system/files/osdi22-han.pdf

E022 / P005 RT end-to-end latency overhead

Field	Evidence
Value	<1.5
Units	percent
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	DISB C-E:multiple BE clients
Interpretation	Authors also state0.1ms; keep inequality and scenario range.
Evidence location	p549 section7.2 Fig11 prose; setup p548 Table2
Source	https://www.usenix.org/system/files/osdi22-han.pdf

E023 / P005 Overall throughput ratio

Field	Evidence
Value	3.00
Units	x
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	DISB D:5 RT model clients20req/s uniform +5 BE model clients
Interpretation	About25% lower throughput than GPUStreams; latency trade-off preserved.
Evidence location	p549 section7.2 Fig11 prose; setup p548 Table2
Source	https://www.usenix.org/system/files/osdi22-han.pdf

E024 / P005 Overall throughput ratio

Field	Evidence
Value	2.96
Units	x
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	DISB E:5 RT model clients20req/s Poisson +5 BE model clients
Interpretation	About25% lower throughput than GPUStreams; latency trade-off preserved.
Evidence location	p549 section7.2 Fig11 prose; setup p548 Table2
Source	`https://www.usenix.org/system/files/osdi22-han.pdf`

E025 / P005 Overall throughput ratio

Field	Evidence
Value	7.7
Units	x
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB
Workload	Apollo REAL trace about43 RTreq/s +5 BE model clients
Interpretation	Real-world trace mapped to representative DNN models; not production deployment measurement.
Evidence location	p549 section7.2 Fig11 prose; setup p548 Table2
Source	`https://www.usenix.org/system/files/osdi22-han.pdf`

E026 / P005 RT end-to-end latency overhead

Field	Evidence
Value	<2
Units	percent
Baseline	RT-Only dedicated GPU
Hardware	AMD MI5016GB

Field	Evidence
Workload	Apollo REAL trace about43 RTreq/s +5 BE model clients
Interpretation	Not a p99 latency claim or security/fault isolation test.
Evidence location	p549 section7.2 Fig11 prose; setup p548 Table2
Source	`https://www.usenix.org/system/files/osdi22-han.pdf`

E027 / P006 ResNet18 p99 reduction factor

Field	Evidence
Value	up to3
Units	x
Baseline	CUDA-MS
Hardware	Tesla T4
Workload	`ResNet18+InceptionV3;lognormal` sigma1.5/2
Interpretation	Author-reported approximate maximum; not uniform across loads
Evidence location	printed p604 section7.1/Fig12; PDF pp10-11
Source	`https://kelvin-ng.github.io/assets/sosp2023-final224.pdf`

E028 / P006 p90 instrumentation overhead

Field	Evidence
Value	6.6
Units	microseconds
Baseline	Uninstrumented kernel
Hardware	Tesla T4
Workload	Empty kernel;160 blocks;instrumentation with aggregation
Interpretation	Microbenchmark; not inference request latency
Evidence location	printed p606 section7.3/Fig15; PDF p12

Field	Evidence
Source	https://kelvin-ng.github.io/assets/sosp2023-final224.pdf

E029 / P007 Aggregate throughput ratio

Field	Evidence
Value	up to7.3
Units	x
Baseline	Dedicated high-priority inference
Hardware	V10016GB;A10040GB extension
Workload	Two inference clients;Poisson;V100
Interpretation	Maximum; includes best-effort work
Evidence location	printed p1086 section6.2.3/Fig12; PDF p12
Source	https://anakli.inf.ethz.ch/papers/orion_eurosys24.pdf

E030 / P007 High-priority p99 overhead

Field	Evidence
Value	15
Units	percent
Baseline	Dedicated high-priority inference
Hardware	V10016GB;A10040GB extension
Workload	Two inference clients;Poisson;V100
Interpretation	Authors summarize within15%; averaged across collocations, not worst-case guarantee
Evidence location	printed p1086 section6.2.3/Fig12; PDF p12
Source	https://anakli.inf.ethz.ch/papers/orion_eurosys24.pdf

E031 / P008 GPU utilization efficacy GUE

Field	Evidence
Value	94.04
Units	composite score
Baseline	Default MPS GUE38.82
Hardware	One V10016GB
Workload	Alexnet+ResNet50;25ms target
Interpretation	Not percent GPU utilization
Evidence location	GSLICE PDF p9; printed p500; Table3
Source	`https://www.cse.iitb.ac.in/~{}puru/courses/spring 2024-25/downloads/gslice.pdf`

E032 / P008 Signed latency improvement

Field	Evidence
Value	-7.52
Units	percent
Baseline	TensorRT server19.02
Hardware	One V10016GB
Workload	`ResNet50;batch1;GPU-ready` batch execution
Interpretation	Negative means worse latency; excludes transport overhead; single-model comparison
Evidence location	GSLICE PDF p10; printed p501; Table4 and section4.3
Source	`https://www.cse.iitb.ac.in/~{}puru/courses/spring 2024-25/downloads/gslice.pdf`

E033 / P009 Clockwork goodput

Field	Evidence
Value	approximately800
Units	requests/second

Field	Evidence
Baseline	INFaaS on nonidentical EC2 setup
Hardware	One V10032GB in section6.1
Workload	15 ResNet50 instances;16 clients/model;500ms SLO
Interpretation	Within-SLO successes; comparator qualitative; not total throughput
Evidence location	USENIX PDF pp10-11; printed pp451-452; Figure5 and section6.1
Source	`https://people.mpi-sws.org/~{}antoinek/documents/20osdi_clockwork.pdf`

E034 / P009 Clockwork request latency

Field	Evidence
Value	approximately300
Units	milliseconds
Baseline	No comparative ratio
Hardware	One V10032GB in section6.1
Workload	15 ResNet50 instances;16 clients/model;500ms SLO
Interpretation	Batch16 scheduling explanation; not a p99 value
Evidence location	USENIX PDF pp10-11; printed pp451-452; Figure5 and section6.1
Source	`https://people.mpi-sws.org/~{}antoinek/documents/20osdi_clockwork.pdf`

E035 / P010 Average P99 inference latency improvement factor, as reported

Field	Evidence
Value	12.6
Units	x
Baseline	TaskSwitch (authors' adapted temporal-switch baseline)

Field	Evidence
Hardware	single NVIDIA Tesla V100 16GB in section6.1; four-V100 host; separate section6.6 single A100 80GB LLM experiment
Workload	section6.1 single-V100 DNN inference+Swin-T; `Light/Heavy/Burst/Skewed/MAF` traces;300s
Interpretation	Author-reported average across tested workloads; not a uniform per-model speedup, and not a comparison to unmodified PipeSwitch.
Evidence location	USENIX PDF p11; printed p1666; Figure10 discussion
Source	`https://www.usenix.org/system/files/atc25-wang-jiali.pdf`

E036 / P011 Reported throughput

Field	Evidence
Value	446
Units	requests/s
Baseline	same configuration with1 process (354requests/s)
Hardware	A10080GB; Amazon p4de.24xlarge hosts with8 GPUs; extracted profiler point concerns one MIG segment
Workload	InceptionV3,1-GPC MIG instance,batch4;3 MPS processes
Interpretation	One local configuration, not fleet savings. Source is arXivv1; final published version not verified.
Evidence location	arXiv v1 PDF p5; sectionIII-B; Figures3-4 on p4
Source	`https://arxiv.org/pdf/2409.14447`

E037 / P011 Reported inference latency; quantile unspecified

Field	Evidence
Value	27
Units	ms
Baseline	same configuration with1 process (11ms)

Field	Evidence
Hardware	A10080GB; Amazon p4de.24xlarge hosts with8 GPUs; extracted profiler point concerns one MIG segment
Workload	InceptionV3,1-GPC MIG instance,batch4;3 MPS processes
Interpretation	Quantile/arrival model not established for this profiler point; do not label p99. Value is specific to the accessed author manuscript.
Evidence location	arXiv v1 PDF p5; sectionIII-B; Figures3-4 on p4
Source	`https://arxiv.org/pdf/2409.14447`

E038 / P012 Reported single-GPU throughput factor

Field	Evidence
Value	>10
Units	x
Baseline	Native early binding at same offered function rate
Hardware	single V10032GB in local Figure8; evaluation worker has4 V10032GB GPUs
Workload	multiple ResNet152 functions on one `V10032GB;10requests/minute/function`
Interpretation	Lower bound reported by authors, not an exact10x; function count varies to saturate memory. No fleet cost claim.
Evidence location	USENIX PDF pp10-11; printed pp605-606; Figure8 and section7.2
Source	`https://www.usenix.org/system/files/atc25-yu.pdf`

E039 / P012 Reported P98 latency upper bound

Field	Evidence
Value	<50
Units	ms
Baseline	Native under matched request-rate setting

Field	Evidence
Hardware	single V10032GB in local Figure8; evaluation worker has4 V10032GB GPUs
Workload	same single-GPU low-rate ResNet152 `test,10-80requests/minute/function`
Interpretation	Bound from text, not a digitized point; default98th percentile from section7 metrics. Applies to this workload.
Evidence location	USENIX PDF pp10-11; printed pp605-606; default P98 metrics and section7.2
Source	`https://www.usenix.org/system/files/atc25-yu.pdf`

E040 / P013 Author-reported average throughput factor

Field	Evidence
Value	6.7
Units	x
Baseline	INFless with documented warm-up/pipelined loading/model sharing
Hardware	V10016GB,80CUs; host has4 GPUs; single-GPU multi-user test in section9.4
Workload	three workflow `types,PyTorch/TensorFlow,Azure-derived` invocation patterns; evaluated V100 setup
Interpretation	Aggregate within-paper result across tested workflows; not universal or a matched cross-paper speedup. Batch/concurrency optimizations contribute.
Evidence location	USENIX PDF p11; printed p68; section9.1; Figure13
Source	`https://www.usenix.org/system/files/atc24-wu-hao.pdf`

E041 / P013 Reported API-forwarding overhead range

Field	Evidence
Value	3-5
Units	microseconds

Field	Evidence
Baseline	direct API path; instrumentation overhead measurement
Hardware	V10016GB,80CUs; host has4 GPUs; single-GPU multi-user test in section9.4
Workload	GPU API forwarding with48concurrent functions on one GPU
Interpretation	Per-API overhead, not end-to-end inference latency; many APIs asynchronous.
Evidence location	USENIX PDF p13; printed p70; section9.3; Figure18
Source	`https://www.usenix.org/system/files/atc24-wu-hao.pdf`

E042 / P014 Reported average QoS violation rate

Field	Evidence
Value	1.23
Units	percent of requests
Baseline	original soloP99 threshold (1%baseline violations by definition)
Hardware	A10040GB for local quota/QoS tests; twoRTX4090 used for separate distributed-training test
Workload	Transformer-XL inference; batch8/16/32. Original rate is half solo maximum; shared test halves that rate again and assigns 50% time; tokens5-80ms.
Interpretation	Average across tested configurations; no claim that worst-case violation is1.23%. Threshold-relative metric, not latency in milliseconds.
Evidence location	USENIX PDF pp12-13; printed pp1013-1014; section5.5; Figure17
Source	`https://www.usenix.org/system/files/atc25-zhang-shulai.pdf`

E043 / P014 Reported kernel module host-memory footprint

Field	Evidence
Value	4.8
Units	MB
Baseline	no module; additional host memory cost
Hardware	A10040GB for local quota/QoS tests; twoRTX4090 used for separate distributed-training test
Workload	kernel-space GPU sharing module on evaluated Linux host
Interpretation	Host memory, not GPU VRAM; separate monitoring thread also required.
Evidence location	USENIX PDF p13; printed p1014; section5.7
Source	`https://www.usenix.org/system/files/atc25-zhang-shulai.pdf`

E044 / P015 Reported high-priorityP99 latency ratio

Field	Evidence
Value	1.07
Units	x standalone
Baseline	standalone high-priorityBert-large execution
Hardware	single NVIDIA GV100 for extracted Triton/Paella case; NPU/ASIC/FPGA results outside extracted GPU outcomes
Workload	twoBert-large clients `ononeGV100;high-priority10req/s,backgroundcontinuous`
Interpretation	Triton+XSched;not a rawmillisecond value. NativeTriton `ratio1.53x;noNPUresultincluded.`
Evidence location	p683 section7.4 case3/Figure15a; PDF p14
Source	`https://www.usenix.org/system/files/osdi25-shen-weihang.pdf`

E045 / P015 Author-reported P99 latency reduction

Field	Evidence
Value	30.0

Field	Evidence
Units	percent
Baseline	vanillaTriton (P99=1.53xstandalone)
Hardware	single NVIDIA GV100 for extracted Triton/Paella case; NPU/ASIC/FPGA results outside extracted GPU outcomes
Workload	same `two-clientTriton/GV100case`
Interpretation	Within-paper comparison `only;author-rounded30.0%,not` a universal bound.
Evidence location	p683 section7.4 case3/Figure15a; PDF p14
Source	`https://www.usenix.org/system/files/osdi25-shen-weihang.pdf`

E046 / P018 SLO-compliant goodput

Field	Evidence
Value	524.29
Units	requests/s
Baseline	Clockwork: 457.24 requests/s
Hardware	Dell PowerEdgeR740; V10032GB; local synthetic co-running experiment and separate four-GPU realistic trace.
Workload	Local three-model load ratio 3:1:2
Interpretation	Local outcome, not four-GPU replay or p99
Evidence location	Published pp. 17-18; Table 3
Source	`https://www.sciencedirect.com/science/article/pii/S0167739X23003187`

E047 / P018 Unfairness ratio

Field	Evidence
Value	1.05
Units	ratio
Baseline	Clockwork: 1.50

Field	Evidence
Hardware	Dell PowerEdgeR740; V10032GB; local synthetic co-running experiment and separate four-GPU realistic trace.
Workload	Local three-model load ratio 3:1:2
Interpretation	Maximum/minimum model slowdown; not security isolation
Evidence location	Published pp. 17-18; Table 3; Eq. 2
Source	`https://www.sciencedirect.com/science/article/pii/S0167739X23003187`

E048 / P019 Pairs meeting twice-solo p99 QoS target

Field	Evidence
Value	30
Units	tested pairs
Baseline	MPS Non-ArkGPU
Hardware	Machine with four V100s; Xeon Platinum8163; CentOS7.9.2009; driver460.106.00; CUDA11.1; cuDNN7.6.5.
Workload	Five inference models paired with six training models
Interpretation	Count of tested pairs, not a universal guarantee
Evidence location	Published pp. 313-319; Section 5.2
Source	`https://doi.org/10.1007/s42514-023-00154-y`

E049 / P019 Low-contention QPS loss

Field	Evidence
Value	about 5
Units	percent
Baseline	Unthrottled co-location
Hardware	Machine with four V100s; Xeon Platinum8163; CentOS7.9.2009; driver460.106.00; CUDA11.1; cuDNN7.6.5.
Workload	Low-contention tested pair settings

Field	Evidence
Interpretation	Regression retained; larger system-level goodput gains are not one-GPU results
Evidence location	Section 5.2 source assessment
Source	`https://doi.org/10.1007/s42514-023-00154-y`

E050 / P020 Effect of 15 percent prediction perturbation

Field	Evidence
Value	8.09
Units	percent makespan increase
Baseline	Unperturbed predictor
Hardware	Two A30s,24GB each; CUDA12.2; driver535.171.04; DCGM3.1.3. Local sections4.2/4.3.1 use one A30. StaCon `profiles2g.12gb+1g.6gb+1g.6gb.`
Workload	Reported prediction sensitivity
Interpretation	Sensitivity result; local JCT and cluster simulation are separate
Evidence location	PDF pp. 11-16; Section 4.3.2
Source	`https://doi.org/10.1007/s10586-025-05284-2`

E051 / P021 Evaluated latency statistic

Field	Evidence
Value	95
Units	percentile
Baseline	AMD unrestricted concurrency labelled MPSDefault
Hardware	One MI50/60CUs; dualEPYC7302/512GB; Ubuntu18.04; kernel5.4; ROCm4.5
Workload	Inference workers and heterogeneous pairs
Interpretation	Statistic definition, not a 95-percent latency improvement
Evidence location	PDF pp. 9-10; Sections V-VI; Table III

Field	Evidence
Source	https://danielwong.org/files/KRISP-HPCA2023.pdf

E052 / P022 Normalized background throughput

Field	Evidence
Value	1.19
Units	ratio
Baseline	TGS: 1.65
Hardware	A10040GB; Ubuntu20.04; CUDA12.2; PyTorch2.2; ONNX1.17
Workload	Ten percent idle-time setting
Interpretation	Foreground protection trades background throughput; compare this setting only
Evidence location	PDF pp. 9-11; Sections 5-5.3; Figure 6
Source	https://arxiv.org/pdf/2410.07381

D Calculated quantities

Calculation	Formula and input	Value / limit
C001 P011	(446/354-1)*100: 446requests/s at3processes;354requests/s at1process;InceptionV3,1-GPC,batch4	25.988700564971758 percent. calculated,not reported;based on accessed author manuscript
C002 P011	27/11: 27ms at3processes;11ms at1process;same1-GPC/batch4configuration	2.4545454545454546 x. calculated,not reported;quantile unspecified;based on accessed author manuscript

E Guideline deliverable crosswalk

Item	Requirement	Artifact	Status / qualification
1	Final SLR report	report/main.pdf (draft)	Draft present; verify before final submission
2	Engineering problem	report section 1	Draft present; verify before final submission
3	Research objectives	report section 1	Draft present; verify before final submission
4	Research questions	report section 2	Draft present; verify before final submission
5	SLR protocol	`../task1/submitted_protocol.pdf`	Existing evidence preserved; limits documented
6	Database justification	report section 3	Draft present; verify before final submission
7	Complete search strings	Search Log / appendix	Draft present; verify before final submission
8	Inclusion/exclusion criteria	report section 3	Draft present; verify before final submission
9	Database-wise results	report section 4	Existing evidence preserved; limits documented
10	Search and screening log	Screening Log / Search Log	Existing evidence preserved; limits documented
11	Duplicate record	Duplicate Removal	Existing evidence preserved; limits documented
12	PRISMA flow	`baseline/PRISMA_2020_FLOW_DIAGRAM.pdf`	Draft present; verify before final submission
13	Quality checklist/scores	QA / QA Reconciliation	Existing evidence preserved; limits documented

Item	Requirement	Artifact	Status / qualification
14	Included primary studies	Studies / Source Register	Draft present; verify before final submission
15	Extraction sheet	Extraction / Evidence	Existing evidence preserved; limits documented
16	Taxonomy	Taxonomy / report section 5	Draft present; verify before final submission
17	Comparative matrix	Comparability / report section 7	Draft present; verify before final submission
18	Trade-offs	report section 8	Draft present; verify before final submission
19	Conflicting evidence	Comparability / report section 9	Draft present; verify before final submission
20	All RQ answers	report section 6	Draft present; verify before final submission
21	Engineering recommendation	Recommendations / report section 10	Draft present; verify before final submission
22	Research gaps	Research Gaps / report section 11	Draft present; verify before final submission
23	Threats to validity	Validity / report section 12	Draft present; verify before final submission
24	References	report references	Draft present; verify before final submission
25	Supporting verification files	logs / baseline / scripts	Draft present; verify before final submission

F Journal benchmark and drafting standard

The organizational benchmark is Hong et al. [25]. Its layer-based taxonomy and comparison structure inform the draft. The current review adds the instructor-required search, selection and appraisal records. Journal standing is contextual and does not certify the draft as publishable. See the separate benchmark note for the dated publisher-metric check.

G Files, timestamps and remaining checks

The Excel workbook preserves the original 14 tabs and adds analytical and audit tabs. The complete 4,075-row screening log and 114 duplicate links remain electronically accessible rather than being expanded into hundreds of print pages. New activity timestamps are in UTC and Pakistan time; historical batch/build values retain their original meaning.

Priority task	**Evidence required**
Confirm whether bounded review with 559 unassessed retained records is acceptable	Instructor direction / dated team scope decision. Do not infer approval from general permission to use AI
Supply InferFair, ArkGPU and Dyna-P PDFs already downloaded	Original PDFs or legitimate accessible source. These three publisher copies were not retrieved
Check headline outcomes against source figures/tables	Report ID, evidence ID, source version, page, correction and actual time. Use 52 evidence rows; do not sign unseen rows
Check companion/version mapping	P005/P023 outcomes and P002/P011/P022 manuscript comparisons. Separate report count from study count
Confirm original search artifacts and group provenance	Exact original exports, search fields/facets and times. Do not reconstruct missing event times from file dates